UPGRADE REPORT

Submitted by **ALEXANDER PAUL HOLMAN**, to the University of Exeter as evidence of suitability to upgrade from the degree program of Masters of Philosophy in Physics to the degree program of Doctor of Philosophy in Physics, **20 JULY 2025**.

I certify that all material in this report which is not my own work has been identified and that any material that has previously been submitted and approved for the award of a degree by this or any other University has been acknowledged.

(Signature) . . . .

ALEXANDER PAUL HOLMAN

Dedication

To Timothy, Hiccup, Digby, Brandy and the Girlies,
and my dearest darling Charlotte.

You all changed my life.

Thank you.

Author’s declaration

Declaration

I declare that the work contained in this report is my own, except where otherwise stated. All sources of information and data have been appropriately acknowledged through citations or references.

This report has not been submitted for any other degree or professional qualification, and it represents original work conducted during my studies Master of Philosophy in Physics at the University of Exeter.

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Abstract

Material interfaces underpin the performance of a wide range of functional devices, from microelectronic transistors to catalytic systems. Their properties often emerge from interfacial effects not captured by bulk characterisation, making the prediction of interface stability a critical and computationally demanding task. While Density Functional Theory (DFT) provides accurate energetics, its high cost limits its applicability to large-scale screening. This study investigates whether machine-learned interatomic potentials (MLPs), specifically the MACE framework, can replicate DFT-derived rankings of interface favourability across a diverse set of semiconductor heterostructures.

Interfaces were generated using the ARTEMIS toolkit across three compositional stages: initial $Si|Ge$ binaries, expanded $Si|Ge$ binaries, and a broader C-Si-Ge-Sn set. Each interface structure was independently evaluated using both DFT and MACE. Energies were compared using rank correlation metrics (Spearman ρ, Kendall τ), absolute error measures (RMSE, MAE), and Top-N overlap to assess rank preservation.

Results indicate that MACE's predictive performance is system-dependent. Rank correlation was weak for silicon-only interfaces but improved markedly for chemically ordered alloy systems, such as Si|Ge. Cross-method evaluations suggested that structural relaxation, rather than energy evaluation, is the primary source of deviation between DFT and MLP predictions.

Future work will focus on training delta-learning models to correct systematic MLP errors using DFT data and on developing a predictive machine-learning interface (MLI) model based on slab features. The role of surface orientation and structural features in determining interface stability will also be examined.

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Chapter 1

Introduction

Material interfaces underpin the performance and reliability of advanced solid-state systems [17]. In semiconductor devices, they govern charge transport, carrier confinement, defect formation, and phase stability; in energy conversion and storage technologies, they regulate ion exchange kinetics, chemical compatibility, and thermal management [10,17,33]. In emerging heterostructures, interfacial interactions induce effects not found in the bulk, such as Schottky barriers or phase stabilisation [32]. As device dimensions shrink and surface-to-volume ratios increase, interfacial effects dominate performance, underscoring the importance of modelling interfaces with quantitative tools [9, 19, 29].

Material interfaces delineate the boundary where two distinct solids meet, whether as heterostructure junctions, grain boundaries, or thin films . These exhibit unique atomic, electronic, and thermal characteristics that set them apart from the bulk phases [17,19]. Local features such as strain fields, atomic reconstruction, and elemental intermixing create modified bonding environments and electronic states, profoundly influencing the mechanical, transport, and optical behaviour of the composite [26,28,33]. In micro- and nanoscale systems, interfaces largely determine electronic and mechanical performance [29].

In this research, material interfaces are studied predominantly through ab initio methods, particularly density functional theory (DFT) and new machine-learned interatomic potentials (MLPs) .

Band offsets at interfaces, often deviating from predictions by simple models like Anderson's rule, are governed by local charge redistribution, dipole formation, and interface-specific states . The emergence of new electronic phases at interfaces, such as Ohmic contact formation or different band alignments, necessitates accurate modelling beyond bulk considerations .

Several methodologies have been developed to address the challenges of constructing realistic interface models . The ARTEMIS tool enables high-throughput generation of interface structures through lattice matching and surface termination selection, allowing systematic exploration of the complex interfacial phase space [54]. Similarly, the RAFFLE method uses statistical learning and void-filling algorithms to construct interface phases that are energetically favourable and representative of plausible morphologies [52].

Interfaces also significantly influence phonon transport [26]. They can reduce thermal conductivity compared to bulk values; an effect exploitable in thermoelectric applications [18]. Interfacial scattering, acoustic impedance mismatch, and atomic-scale disorder all contribute to this behaviour, reinforcing the interface's role as a functional component rather than a passive boundary [22]. Thus, in the context of this research, a material interface is not merely a geometric boundary but an emergent, functionally distinct region . It governs critical physical properties, demands precise structural modelling, and enables advances in material innovation and device engineering .

Material interfaces can be broadly categorised by their crystallographic orientation, dimensionality, and degree of lattice coherence . These classifications are instrumental in understanding interfacial phenomena and in designing simulation protocols that are physically realistic and computationally tractable [17, 19, 29].

Interfaces can also be distinguished by the degree of lattice match across the boundary:

- Coherent interfaces exhibit a perfect registry between atomic planes, typically occurring when lattice mismatch is negligible . These interfaces preserve the translational symmetry across the junction and are often found in homoepitaxial or pseudomorphic systems [13].

- Semi-coherent interfaces accommodate lattice mismatch through periodic misfit dislocations . These dislocations relieve interfacial strain while maintaining partial registry, allowing larger domains of coherency interspersed with localised defects [13].

- Incoherent interfaces display no periodicity match across the interface, leading to structurally disordered junctions . These are often found in polycrystalline or amorphous/crystalline composites and can act as strong phonon or electron scattering centres [13].

Chemical composition also plays a contrasting and defining role .

- Heterojunctions involve distinct materials with differing electronegativity, band gaps, and ionic/covalent character . Heterojunctions are central to semiconductor devices, solar cells, and catalysis, often giving rise to emergent interfacial dipoles, states, or reconstructed bonding motifs .

  A heterostructure is a structure composed of multiple heterojunctions .

- Homojunctions are interfaces within a single material where the crystal lattice remains continuous, but local properties differ, such as variations in dopant concentration or applied strain, resulting in regions with distinct electronic or mechanical behaviour .

> As with heterojunction vs structure, homostructure is a structure composed of multiple homojunctions .

These effects are particularly pronounced in systems involving 2D materials, complex oxides, or strongly correlated electron systems [28].

Interfaces often host localised states arising from atomic mismatch, bonding reconstruction, or impurity segregation . These can act as traps or recombination centres, significantly affecting carrier's lifetimes and mobility [13, 22, 33].

The dielectric properties of materials can be strongly altered at interfaces . In systems like $\mathrm{CaCu_3Ti_4O_{12}}$, colossal permittivity arises not from the bulk crystal but from insulating grain boundaries and a semiconducting interior, exemplifying the Maxwell-Wagner effect [53]. Interface-induced polarisation, charge accumulation, and ionic displacements all contribute to enhanced dielectric responses [29, 32].

Thermal transport across interfaces is typically suppressed due to phonon scattering, mass disorder, and acoustic impedance mismatch [30]. This results in thermal boundary resistance (Kapitza resistance), which strongly influences heat flow in thermoelectric and microelectronic devices [22,26,29]. In heterostructures, the interface can significantly scatter phonons, reducing thermal conductivity below that of either constituent [13, 17].

Interfaces can stabilise new or metastable phases not accessible in bulk [45, 52]. For instance, the interface between $\mathrm{BaTiO_3}$ and $\mathrm{SiO_2}$ has been shown to promote the spontaneous formation of the fresnoite phase $\mathrm{Ba_2TiSi_2O_8}$, altering both electronic and mechanical response. A more nuanced example is the $graphene|\mathrm{MgO}$ interface: while monolayer MgO confined between graphene layers often adopts a rocksalt-like structure, this is not simply a case of one phase being energetically favoured. Instead, the local environment, including charge transfer and encapsulation, stabilises the rocksalt phase relative to the expected hexagonal phase in unsupported monolayers. However, experimental and theoretical studies suggest that a mixed or hybrid phase may also emerge with phase stability dependent on thickness and interface strain [39].

Mechanical properties such as bulk modulus, shear strength, and adhesion energy are modified at interfaces due to altered bonding environments and lattice mismatch . These variations are important in the design of composite materials, flexible electronics, and nanoindentation studies [13, 29, 60].

Interfaces affect the optical absorption spectrum by introducing new electronic states and modifying the joint density of states . Strain-induced effects at the interface can shift bandgap and optical properties, which are useful for photodetectors and photovoltaic devices [11, 32].

Accurate control and prediction of material interfaces are vital for the design of next-generation electronic, optoelectronic, and thermal devices [32].

Yet despite their importance, predicting which interfaces are energetically favourable remains a significant challenge . The structural space is vast: each pair of surfaces can be stacked with arbitrary lateral shifts, terminations, and alignments, resulting in a combinatorial explosion of configurations [17, 19, 60]. Strain relaxation, interlayer interactions, and charge transfer further complicate the energy landscape . Accurately assessing favourability across this space demands a balance between resolution and scale that conventional methods struggle to achieve [15, 19, 32].

First-principles techniques such as DFT remain the standard for computing accurate interfacial energies and relaxations [10]. However, their high computational cost, scaling cubically $O(n^3)$ with system size, renders exhaustive sampling intractable for realistic sizes structures, which often comprise hundreds to thousands of atoms [9, 43, 46]. This bottleneck precludes broad screening efforts, especially when exploring multiple chemistries, orientations, and stacking arrangements . As such, DFT might be better suited to final validation and a confirmation of datasets evaluated in a less computationally expensive manner .

The present project explores the use of MLPs for scalable interface screening . MLPs are data-driven models trained to reproduce DFT-level energies and forces, but at orders of magnitude lower cost [57]. In particular, this work focuses on the MACE framework; an equivariant message-passing architecture that preserves rotational symmetries and captures local atomic interactions with high fidelity . By using MACE to rapidly evaluate large numbers of candidate interfaces, it becomes possible to rank structures by predicted stability and prioritise low energetic configurations for subsequent DFT validation [5].

This approach supports a hierarchical modelling workflow in which DFT and MLPs are used in tandem: MLPs for breadth, and DFT for depth . It enables rigorous testing of whether MLPs can replicate DFT-derived favourability rankings across a range of systems, including phase boundaries (e.g. $\mathrm{Si|Si}$) and heterostructures (e.g. $\mathrm{Si|Ge}$, $\mathrm{Si|C}$) . Moreover, it lays the foundation for further improvements through techniques such as delta-learning, structure-based prediction models, and surface-to-interface generalisation .

The remainder of this report is structured as follows :

- **Chapter 2** reviews the theoretical and computational background for interface modelling . It defines material interfaces, explores interfacial energetics and physical effects, and surveys applications across electronic, catalytic, and energy systems . The chapter also introduces DFT, MLPs, and recent algorithmic tools for generating and evaluating heterostructures .
- **Chapter 3** evaluates the use of MACE as a surrogate model for efficiently screening interface structures prior to DFT . It benchmarks MACE vs DFT energetics across over roughly 680 heterostructures, analyses rank correlations, and identifies system-specific performance trends that inform future dataset construction for predictive modelling .

- **Chapter 4** outlines the design and evaluation of a Machine-Learned Interface (MLI) model trained to predict interface favourability . It compares three training strategies baseline MACE, interface-specific fine-tuning, and delta-learning correction, and introduces ranking metrics to quantify predictive performance . The chapter also explores how structural asymmetries, surface orientation, and chemical diversity impact model generalisation .

- **Chapter 5** reflects on key findings across materials systems, highlighting the strengths and limitations of current MLPs in replicating DFT interface rankings .

Through this integrated modelling framework, the project seeks to reduce the cost and uncertainty of interface prediction while preserving the fidelity needed for scientific insight and technological relevance .

1.1 Motivation for the Research

Interfaces rarely form under ideal conditions . Instead, they emerge through complex, often non-equilibrium processes such as high-temperature annealing, epitaxial growth, and chemically driven surface reconstructions . These dynamics frequently trap systems in metastable configurations or generate local disorder, resulting in atomic arrangements that deviate significantly from those predicted by ideal lattice-matching rules . Such complexity challenges efforts to model interface behaviour using conventional approaches alone [17, 19, 22, 29].

While DFT provides a reliable route to total energy prediction and structural relaxation, its unfavourable scaling with system size ($\mathcal{O}(N^3)$ in electron count) severely limits its use in high-throughput workflows . In particular, DFT becomes prohibitive when applied to interface structures exceeding a few hundred atoms, as required to capture long-range strain, multi-layer reconstruction, or compositional variation [9, 19]. Moreover, standard DFT approximations, such as LDA exchange-correlation functionals, often struggle to predict key interfacial phenomena in layered systems . For example, accurate band-alignment at a heterojunction determines charge-injection barriers and device turn-on voltages; and weak van der Waals interactions between two-dimensional layers govern adhesion, interlayer spacing, and electronic reconstructions . Failure to capture these effects can lead to qualitatively incorrect predictions of interface stability and function and requires a higher level of DFT [19, 32].

MLPs offer a promising route to overcoming these limitations . By learning to emulate DFT-level energies and forces, MLPs like MACE enable large-scale structure relaxations and energy evaluations at dramatically reduced computational cost [26, 57]. This opens up the exploration of far broader configurational spaces, including low-symmetry interfaces and extended defects, that would otherwise be intractable using DFT alone . However, this benefit is only realised if the models generalise well across diverse interfacial environments . Achieving DFT-comparable accuracy in such settings requires a representative, high-fidelity training set: any gaps or systematic biases in the reference data will be reflected in the potential's predictions [6, 15, 48].

Chapter 3 shows that MLPs often preserve relative DFT energy rankings, particularly within chemically similar systems, with a few edge cases like pure-Si interfaces (grain boundaries) . These discrepancies underscore the need for targeted refinement, motivating the delta-learning approaches presented in Chapter 4, which correct systematic MLP-DFT residuals via supervised learning on interface-specific error distributions .

Ultimately, this project is motivated by the need for more efficient and predictive tools to accelerate the design of functional heterostructures . By embedding MLPs within automated screening workflows and augmenting them with delta-learning corrections and structure-based heuristics, it becomes possible to bridge the gap between physical realism and computational tractability . In doing so, this research contributes to the broader aim of enabling scalable, data-driven exploration of complex interfacial systems; particularly in regimes where bulk-derived heuristics break down .

1.2 Research Objectives

This PhD aims to develop a Machine-Learned Interface (MLI) model that, given two input materials; whether slabs, bulks, or combinations, can generate realistic and physically plausible interface structures between them . The purpose of this model is not to directly predict interfacial energies or properties, but to output candidate geometries that are likely to be realisable and energetically reasonable, enabling downstream evaluation via existing MLPs or DFT .

The ultimate goal is to accelerate interface discovery by replacing brute-force combinatorial stacking with an intelligent generative model . By learning patterns in crystallography, registry, and bonding across known interfaces, the MLI model will propose viable configurations that reduce the need for exhaustive enumeration and manual filtering .

Such a tool could eventually support applications beyond interface prediction, including surface termination prediction under reactive gas atmospheres, simulation of crystal growth through layer-by-layer stacking, and real-time assessment of growth defects during slab assembly .

To support development of the MLI model, the first phase of this research focuses on constructing a large, diverse dataset of heterostructure interfaces . Since direct generation and relaxation of thousands of interfaces using DFT would be prohibitively expensive, machine-learned interatomic potentials (MLPs), specifically the MACE framework, are evaluated for their ability to replicate DFT-derived rankings of interface favourability [5].

If MLPs can reliably identify promising structures, even without full relaxation, they can serve as a low-cost screening tool to filter candidate interfaces for inclusion in the training set .

This enables the creation of a scalable data pipeline where realistic interfaces are generated via tools like ARTEMIS [54], screened using MLPs, and selectively validated with DFT .

Once promising interfaces are identified, a subset will be fully relaxed using DFT to provide high-fidelity data for MLI training . These DFT-relaxed structures will serve as the reference for extracting physically meaningful descriptors, such as surface orientation, lattice mismatch, interfacial strain, atomic registry, and coordination environments, that can be used as features for the MLI model [21].

Although initial results show strong rank correlation between MLP and DFT energies (e.g., $\rho \approx 0.98$ for Sn|Ge), this agreement may degrade in more complex systems or larger interface models. Full DFT validation remains essential to calibrate and benchmark the workflow, particularly for low-symmetry or poorly coordinated interfaces where MLP generalisation may falter.

To further extend this validation, rankings based on MLP-relaxed geometries will be compared to those from DFT-relaxed structures . If similar rank orderings are recovered across both relaxation methods, it would support the use of MLPs for handling much larger and more realistic interfacial systems, ones that are computationally infeasible to relax using DFT alone . This opens the possibility of generating structurally diverse, high-volume datasets beyond the reach of traditional ab initio approaches .

A longer-term extension of this research, contingent on the success of the main MLI model and access to further computational resources, is the development of an inverse interface design framework . Rather than proposing interfaces from input slabs alone, this model would take target properties or desired interfacial features, such as low strain, specific registry types, or stability thresholds, and suggest plausible $A|B$ configurations that meet those criteria .

Such a tool could support experimental design by guiding researchers toward synthesizable interfaces that meet functional requirements . However, training such a model would require a much broader dataset, including properties computed from full DFT relaxations on MLI-generated structures, something only feasible with additional funding and computational time .

To improve the accuracy of interface energy predictions, particularly for systems where MACE performs poorly (e.g. strained silicon-rich structures or cross-material environments), this project will explore refinement strategies for the MLP itself . One approach is delta-learning: training a secondary model to correct systematic energy differences between MACE and DFT, using DFT-relaxed structures as a reference [40].

Another strategy is direct fine-tuning or retraining of MACE on an expanded dataset that includes interface-specific configurations, such as strained slabs, low-symmetry orientations, or cross-material bonding . This is motivated by known failure modes in MLIPs, including poor generalisation to novel bonding environments, and underprediction of surface and interface energies [29, 56]. These efforts will aim to mitigate known risks; such as extrapolation errors, by enriching the training distribution with interfacial scenarios that challenge model stability . By explicitly introducing diverse bonding motifs and strain conditions, the model's predictive performance can be made more robust across real-world interface configurations, reducing the likelihood of catastrophic errors in high-throughput screening .

A further objective of this research is to identify and mitigate failure modes in machine-learned potentials when applied to interfacial systems .

Despite strong average performance, modern MLIPs such as MACE exhibit known limitations, including potential energy surface (PES) softening, poor extrapolation to strained or defective configurations, and underprediction of surface energies [6].

This work will systematically assess when and why these failures occur, starting with Group IV interfaces as test cases. Specific attention will be paid to bonding asymmetries, coordination anomalies, and cases where poor generalisation leads to incorrect rank ordering or unrealistic relaxations. Where needed, refinement strategies such as delta-learning, fine-tuning, and active learning will be deployed to correct these behaviours [40]. The goal is not only to improve model accuracy, but also to build a deeper understanding of where and why MLPs break down when applied to out-of-distribution geometries; informing the development of more robust, uncertainty-aware workflows for interface prediction .

In parallel with predictive modelling, the project seeks to uncover the structural and chemical features that determine interfacial stability . One key line of enquiry is whether surface favourability correlates with interface favourability; for example, whether low-energy surfaces tend to form stable interfaces, or whether compensating effects between unfavourable surfaces can yield unexpectedly favourable interfaces .

Further investigations will explore how stability varies across crystal types (e.g. fcc, hexagonal, tetragonal), material classes (e.g. semiconductors, metals, insulators), and bonding mechanisms (e.g. covalent, ionic, metallic) . The influence of valence orbitals; such as s versus p dominance, will also be considered, particularly in how they affect registry, relaxation, and bonding across the interface .

Together, these objectives form a cohesive research trajectory: from benchmarking machine-learned potentials and constructing curated datasets, to training a generative interface model (MLI) and exploring its extension toward inverse design . The project aims not only to reduce the computational cost of interface exploration, but also to uncover structural principles and predictive strategies that generalise across materials systems .

By integrating machine learning (ML) with crystallographic modelling and ab initio validation, this work supports the development of practical tools for scalable, data-driven interface discovery in electronic, optoelectronic, and catalytic materials .

1.3 Scientific and Practical Significance

Material interfaces are central to the behaviour of functional solids, especially at the nanoscale where surface and interfacial effects dominate bulk properties [10, 11, 26, 28]. In semiconductors, they define charge transport pathways by setting potential barriers and band offsets [32]. In batteries, they regulate ionic conductivity, chemical stability, and degradation kinetics [29]. In photovoltaics and quantum devices, interfacial dipoles, tunnelling channels, and confinement effects underpin key device operations [10, 32].

Subtle differences in stacking, coherence, or atomic reconstruction can shift critical electronic properties by several electronvolts; altering conductivity, carrier lifetimes, or activation barriers . These shifts emerge from a complex interplay of interfacial strain, chemical bonding, and electronic redistribution, and are sensitive to both global structure and local environment [6, 13, 19]. As such, predicting interface stability and functionality from atomic configuration alone remains a demanding task .

While DFT offers a robust ab initio framework for computing interfacial energies and electronic properties, its computational cost scales poorly with system size [10, 29, 46]. Realistic supercells, often comprising hundreds of atoms, are therefore difficult to study in large numbers . DFT is well-suited to final validation, but ill-suited to exhaustive exploration across configurational space .

Machine learning models, particularly MLPs, provide a way forward . When trained on DFT data, MLPs can reproduce formation energies and atomic forces at orders-of-magnitude lower cost, making them suitable for broad screening applications . Their integration into automated workflows, using tools such as ARTEMIS and RAFFLE, enables systematic generation and evaluation of interface candidates across crystallographic orientations, stacking shifts, and chemical combinations [51, 52].

Importantly, the utility of ML does not stop at static energetics . Many applications require insight into how interfaces behave under external stimuli; electric fields, temperature gradients, or chemical exposure [9, 11, 26, 29]. More adaptive and scalable modelling approaches are required for emerging interfacial phases, such as field-tunable dielectric layers or metastable heterostructures .

This project contributes to that broader effort. It assesses the ability of MLPs, specifically MACE, to recover DFT-derived favourability rankings, identifies where discrepancies arise, and proposes delta-learning corrections to address them . It also lays groundwork for predictive models that estimate interface properties from top and bottom slabs, enabling pre-screening before full interface construction . By focusing on group-IV semiconductor systems and both homostructure and heterostructure cases, this work targets a class of technologically relevant interfaces where modelling uncertainty remains high and conventional rules often fail .

Ultimately, this research supports the advancement of data-driven, scalable methods for interface discovery; bridging the gap between computational feasibility and the physical fidelity required for next-generation materials design .

Chapter 2

Background and Literature Review

2.1 Material Interfaces

Building on our foundational definitions, we now explore the key functional regions of a material interface, the engineering properties that govern strength, adhesion, and stability, and the principal physical effects arising at these junctions, electronic phenomena and energy-transport processes . We then discuss design parameters for tailoring interfacial behaviour, the role of crystallographic orientation and polymorph-driven reconstruction, and the generation and prediction of interfacial defects .

In materials engineering, interfaces govern a wide array of properties including adhesion, corrosion resistance, mechanical strength, and diffusion kinetics [9,28]. The strength and ductility of composite systems or coatings often depend on interfacial bonding and lattice compatibility [31, 42]. Tailoring interfacial chemistry or morphology, through methods such as surface functionalisation, alloying, or atomic layer deposition, is often essential for achieving desired macroscopic performance metrics [9, 11, 28].

Semiconductor heterojunctions, metal-semiconductor contacts, and dielectric boundaries are fundamental to the operation of transistors, diodes, and capacitors [26, 32]. Localised changes in atomic registry, roughness, or defect density at the interface can significantly alter tunnelling behaviour, barrier heights, and mobility [19, 26]. Where interfacial dipoles, charge redistribution, and electrostatic reconstruction result in complex electronic responses not captured by bulk models [2].

Energy systems are similarly interface-dominated. In lithium-ion batteries, for example, the solid-electrolyte interphase (SEI) acts as a metastable interfacial layer that dictates both stability and charge transport [9]. The performance and longevity of solid-state batteries, fuel cells, and thermoelectric devices can depend on the engineered properties of interfaces, which affect ionic conductivity, catalytic activity, thermal transport, and electronic insulation [28]. In photovoltaics, interfacial band offsets between donor and acceptor layers, or between absorber materials and charge transport layers, govern charge separation efficiency and open-circuit voltage [32].

Overall, interfaces are not incidental features but active design parameters . As such, mastering interfacial science is a gateway to developing high-performance materials and devices .

The structural features that influence interface behaviour include crystallographic orientation, the nature of the phase boundary, and the presence of defects [60].

The relative orientation of crystals on either side of an interface can significantly affect the atomic coordination, bond continuity, and symmetry properties across the interface [27, 60]. Epitaxial alignment, where lattice planes across the interface maintain a well-defined relationship, typically results in low-strain, coherent interfaces [13]. In contrast, a mismatch in orientation can lead to misfit dislocations or interfacial strain, influencing charge carrier transport, interfacial states, and mechanical stability [19,28,29]. Orientation also affects the formation energy and electronic reconstruction at the interface, which can lead to phenomena like interface-induced states or altered band alignments [2, 11, 60].

2.2 Constructing Interfaces: Lattices, Orientations, and Matching Algorithms

Lattices and Unit Cells

In crystalline solids, a lattice is defined as a periodic array of points generated by integer translations of three fundamental vectors, known as lattice vectors $(\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3)$. Mathematically, any lattice point $\mathbf{R}$ is expressed as [20]:

$$\mathbf{R} = n_1\mathbf{a}_1 + n_2\mathbf{a}_2 + n_3\mathbf{a}_3, \quad n_i \in \mathbb{Z}. \tag{2.1}$$

The choice of lattice vectors (which need not be orthogonal) defines the geometry and symmetry of the unit cell, the smallest repeating structural unit of the crystal . A given crystal structure can often be described by multiple equivalent representations . Silicon (Si), for instance, crystallizes in a diamond-cubic structure and can be represented either as :

- A primitive cell containing 2 atoms with non-orthogonal basis vectors .
- A conventional cubic cell containing 8 atoms, exhibiting face-centered cubic symmetry and belonging to space .
- Group $Fd\bar{3}m$ C.5.2 (Pearson symbol $cF8$ C.3.2) .

Both descriptions are equivalent physically, the primitive cell is smaller (fewer atoms, lower symmetry), while the conventional cell is larger and highlights the cubic symmetry .

The first 4 $groupIV$ elements; carbon (C), silicon (Si), germanium (Ge), and alpha-tin ($\alpha - Sn$), adopt the diamond-cubic structure but differ in lattice constants . For instance, silicon's lattice constant is approximately 5.431 Å, whereas germanium's is around 5.658 Å . These differences create intrinsic lattice mismatches at heterointerfaces, such as a simple $1{:}1$ $Si|Ge$ interface, resulting in a lattice mismatch of roughly :

$$\epsilon = \frac{a_{\text{Ge}} - a_{\text{Si}}}{a_{\text{Si}}} \approx 4.17\%, \tag{2.2}$$

meaning germanium's lattice is about 4.2% larger than silicon's . Bonding these crystals directly forces one side to stretch (under tension) or compress, inducing strain and consequently increasing the total energy due to lattice misfit . Significant mismatches can also lead to the formation of misfit defects, such as dislocations, if the strain is not elastically accommodated [51].

Lattice Mismatch and Supercells

To alleviate strain caused by lattice mismatch at interfaces, one can introduce supercells, enlarged periodic cells formed by integer expansions (n_a, n_b, n_c) of the original lattice vectors . A supercell constructed this way contains $n_a \times n_b \times n_c$ times more atoms than the primitive cell, and its total energy (in the absence of any interface) scales linearly as :

$$E_{\text{supercell}} = E_{\text{unit cell}} \times n_a \times n_b \times n_c. \tag{2.3}$$

For interfaces, the effective mismatch in supercells decreases substantially . For instance, a Si|Ge $22{:}21$ supercell reduces lattice mismatch to :

$$\epsilon_{\text{eff}} = \frac{22a_{\text{Si}} - 21a_{\text{Ge}}}{22a_{\text{Si}}} \approx 0.016\%, \tag{2.4}$$

resulting in significantly lower strain energy [52].

In a homogeneous bulk crystal, using a larger supercell does not change the per-atom energy (since it's just a repetition of the same structure) . However, for a heterogeneous interface, choosing an appropriate common supercell can distribute the mismatch strain over a larger period, thereby reducing the strain per unit length and stabilising the interface . Essentially, we seek integer multiples of the two lattice constants that coincide as closely as possible . For example, repeating the Si lattice 25 times and the Ge lattice 24 times yields $25a_{\mathrm{Si}} \approx 24a_{\mathrm{Ge}}$, leaving an effective mismatch below 0.02% . Such a $25{:}24$ commensurate supercell drastically minimizes the residual strain energy compared to the original 4% mismatch . In practice, researchers often use this approach (sometimes called a Moiré supercell in 2D contexts) to reduce mismatches to under 1% or less, at which point the interface can be nearly strain-free .

It should be noted that making the supercell very large will reduce strain but at the cost of more atoms in the simulation . Thus, there is a trade-off: we want the smallest supercell that brings the lattices into acceptable registry . As the supercell size grows, the interfacial energy (per area) will converge to a stable value because the misfit strain is distributed over more atoms, and the system can relax closer to bulk-like conditions .

Orientation and Planar Matching

Besides matching lattice constants, crystallographic orientation plays a crucial role in constructing realistic interfaces . Crystal orientations are commonly described by Miller indices (hkl), which specify the orientation of a crystallographic plane relative to the unit cell axes . For example, a (100) surface represents a cut perpendicular to the x-axis of a cubic cell, a (110) surface is a diagonal cut, and a (111) plane slices diagonally across the cubic cell . When constructing an interface, one must carefully select which Miller-indexed plane of material A will contact a particular plane of material B . Rotating and aligning slabs according to these indices determine the atomic registry at the interface, significantly affecting interfacial properties such as bonding patterns, strain distribution, and energetics [51].

Different Miller planes offer distinct surface terminations, each with unique atomic configurations . Thus, even a single crystallographic orientation may yield several energetically and structurally distinct interfaces depending on the chosen termination . Moreover, some seemingly mismatched orientations can be rendered commensurate by careful in-plane rotation . For instance, high-throughput searches have identified viable alignments between dissimilar crystal planes (e.g. cubic (100) against hexagonal-like (111)), minimising residual mismatch through rotation .

Additionally, crystals with differing stacking sequences can form interfaces: a face-centered cubic $(FCC)(111)$ surface has ABC atomic stacking, whereas a hexagonal close-packed $(HCP)(0001)$surface follows $ABAB$ stacking . Joining these surfaces can lead to multiple possible stacking registries (atomic alignments across the interface), each with unique structural and energetic characteristics .

Interface registry involves two primary degrees of freedom :

- In-plane shifts ($\mathbf{d}_{\|}$): lateral translations within the interface plane, altering atomic overlap and influencing bonding arrangements .
- Out-of-plane separations ($\mathbf{d}_{\perp}$): vertical adjustments controlling bonding distances between slabs, or introducing vacuum gaps to simulate weakly bonded or non-bonded interfaces .

All parameters above, together, define the interfacial configuration space, requiring systematic exploration to identify optimal interface structures .

Automated Interface Construction (ARTEMIS)

To systematically explore the large configurational space involved in constructing interfaces and identify physically plausible, low-strain structures, automated algorithms like ARTEMIS are employed [51]. ARTEMIS automates the construction of chemically realistic interfaces by following a structured, step-by-step workflow (Figure 2.5):

1. Surface selection: The user specifies Miller indices $(hkl)_A, (h'k'l')_B$ for surfaces of materials A and B, typically choosing low-energy or experimentally relevant facets .

2. Slab generation: Each bulk crystal is cleaved along the chosen Miller planes, producing slabs with sufficient vacuum to avoid interactions between periodic images and ensuring stoichiometric and chemically reasonable terminations . Multiple possible surface terminations can be considered if relevant .

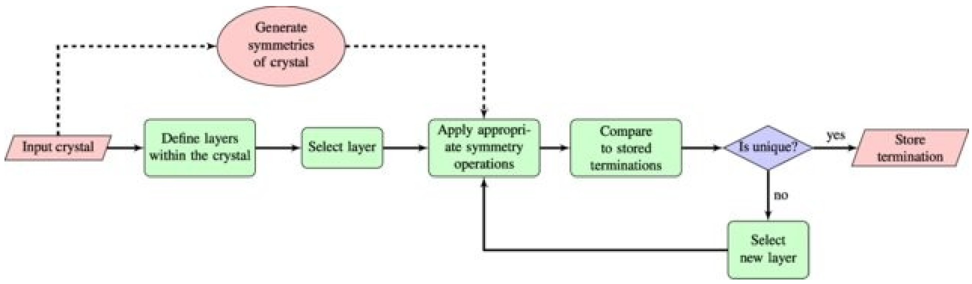

Figure 2.1: Workflow for identifying unique surface terminations in ARTEMIS. Starting from a user-specified crystal and Miller index, candidate slabs are generated by cleaving the structure layer by layer. Each termination is symmetry-checked against previously stored surfaces to ensure uniqueness before storage. This process enables enumeration of all symmetry-distinct terminations for a given surface orientation. Used with permission from Taylor et al. [51].

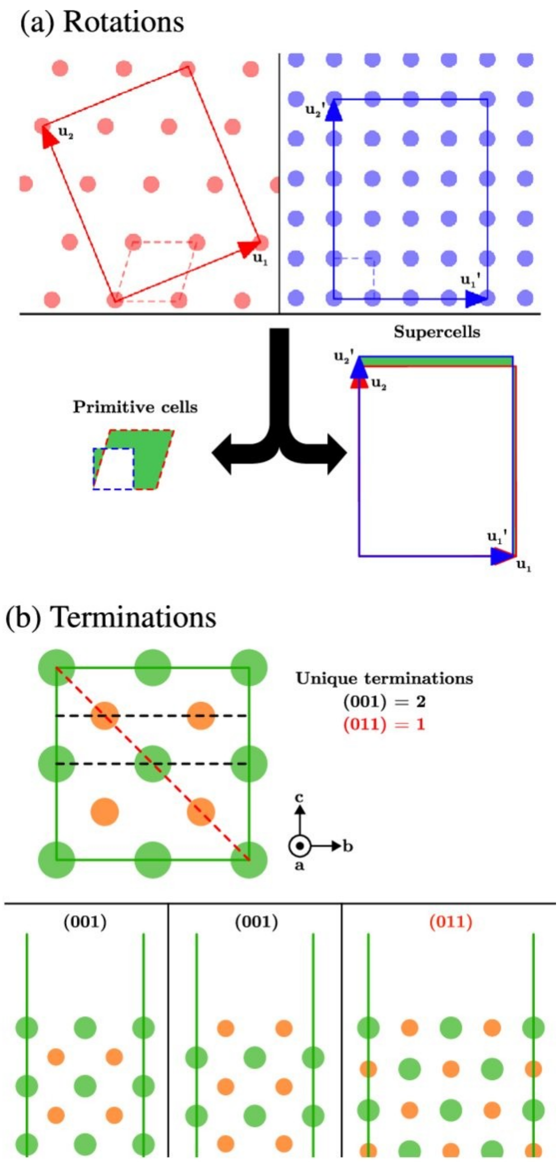

Figure 2.2: Schematic overview of surface matching and termination enumeration. (a) Lattice rotations and supercells: Each material surface is rotated and rescaled to identify supercell combinations with minimal mismatch. Primitive cells (dashed lines) are expanded into larger supercells (solid lines) to improve geometric alignment across the interface. (b) Surface terminations: Cleaving along a given Miller plane can yield multiple unique surface terminations. The (001) orientation allows two distinct terminations, while the (011) yields only one, based on atomic layer stacking along the cleavage direction. Used with permission from Taylor et al. [51].

3. 2D lattice matching: ARTEMIS employs a lattice-matching algorithm based on the Zur-McGill method, seeking integer matrices M_A and M_B that closely approximate the two materials' lattice vectors :

$$M_A[\mathbf{a}_1^A, \mathbf{a}_2^A] \approx M_B[\mathbf{a}_1^B, \mathbf{a}_2^B], \tag{2.5}$$

thereby minimizing lattice mismatch strain [51,63]. Candidate supercells (integer multiples of slab lattice vectors) are generated and compared to find commensurate matches within a small strain tolerance (typically $< 5\%$) . Slight rotations can also be explored at this stage to further reduce mismatch .

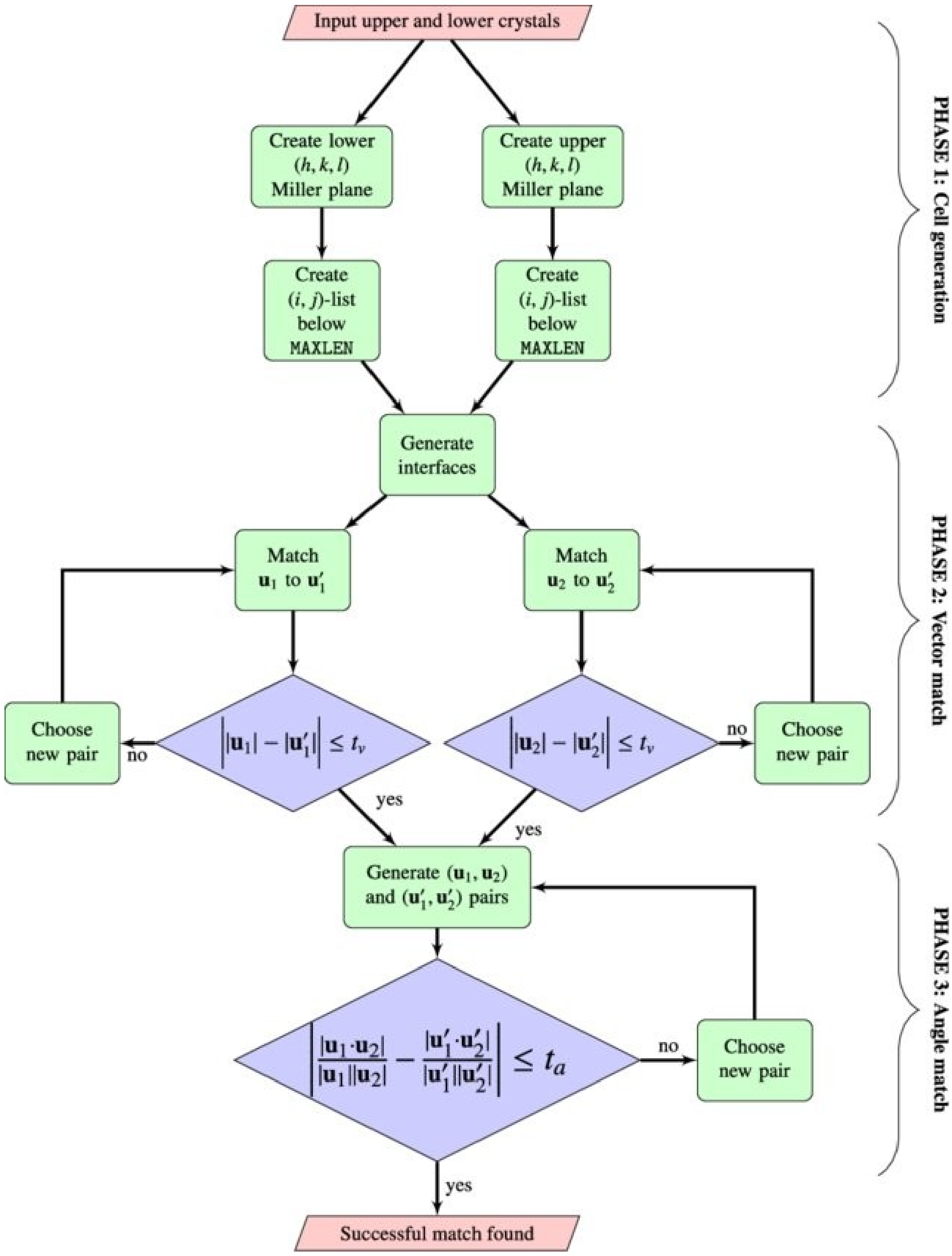

Figure 2.3: Lattice matching workflow used in ARTEMIS, based on the Zur–McGill algorithm. The process is divided into three phases: Phase 1 (Cell generation): Integer supercell candidates are generated from specified Miller planes $(hkl)_A$ and $(h'k'l')_B$, constrained by a maximum length parameter. Phase 2 (Vector match): Pairs of in-plane lattice vectors $(\mathbf{u}_1, \mathbf{u}_2)$ and $(\mathbf{u}'_1, \mathbf{u}'_2)$ are compared, accepting only those whose magnitudes differ by less than a vector tolerance t_v. Phase 3 (Angle match): Accepted vector pairs are filtered by angular mismatch, retaining only those satisfying the angle tolerance t_a. If all criteria are met, a successful lattice match is declared. Used with permission from Taylor et al. [51].

4. Registry enumeration (in-plane shifts): Given a commensurate lattice match, ARTEMIS systematically explores multiple lateral translations ($\mathbf{d}_{\parallel}$) of one slab relative to the other . Fractional offsets (grid sampling) ensure that all high-symmetry and distinct stacking configurations are sampled, significantly influencing the resulting atomic registry and bonding environment .

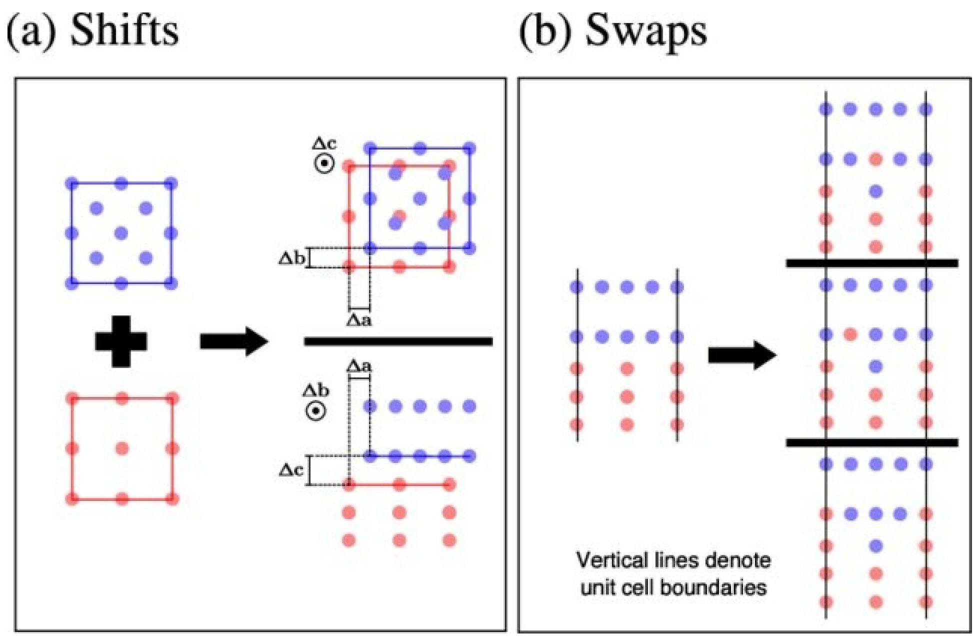

Figure 2.4: The top and bottom crystal slabs are laterally displaced relative to each other to sample different in-plane (Δa, Δb) and out-of-plane (Δc) alignments. Top right shows shifts applied in the interface plane; bottom right shows shifts along the stacking direction. (b) Swaps: Atomic layers near the interface are interchanged across the boundary to introduce intermixing. The left shows the initial unmixed interface. The right shows three distinct swap configurations (top, middle, bottom), separated by bold horizontal lines. Vertical black lines indicate periodic unit cell boundaries; atoms on a boundary are mirrored across the opposite edge. Used with permission from Taylor et al. [51].

5. Vertical optimization (out-of-plane separation): For each in-plane shifted configuration, the perpendicular slab separation ($\mathbf{d}_{\perp}$) is adjusted to ensure realistic bonding distances . Slabs are moved toward each other until physically meaningful interatomic distances or bonding criteria are satisfied, avoiding atomic overlaps or excessive gaps .

6. Filtering candidates: To manage computational complexity, ARTEMIS filters interface candidates based on :

   - Maximum allowable supercell size (limiting the total atom count) .
   - Maximum strain thresholds (usually below 5%) to ensure structural plausibility .
   - Chemical feasibility, such as minimizing dangling or unsatisfied bonds at the interface .

7. Output generation: Finally, ARTEMIS outputs the most feasible interface candidates as initial structures for further relaxation and analysis via DFT or MLPs . Outputs include metadata such as applied strain, shift vectors, and Miller indices to facilitate subsequent identification and tracking .

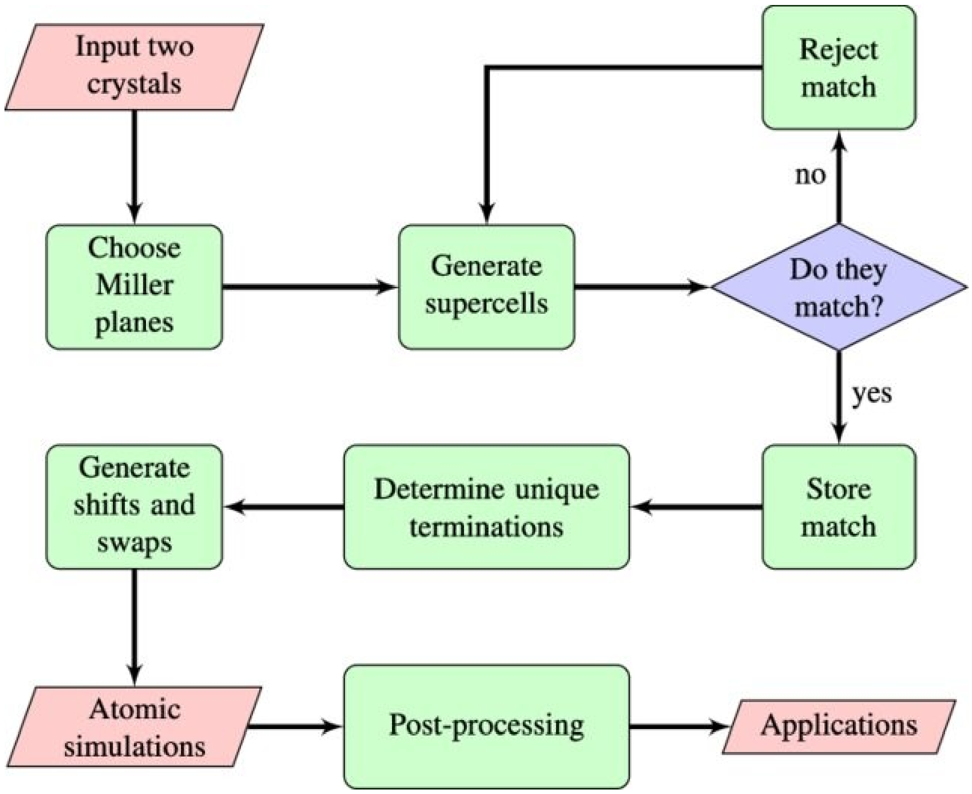

Figure 2.5: End-to-end workflow of the ARTEMIS algorithm for interface construction. Beginning with user-specified Miller planes for two input crystals, ARTEMIS generates and screens supercell combinations for commensurability. Matched surfaces are stored and used to enumerate symmetry-distinct surface terminations. These are further modified via in-plane shifts and interfacial swaps to explore registry and intermixing configurations. Final structures undergo simulation, post-processing, and downstream analysis. This unified workflow enables automated generation of physically plausible, chemically diverse interface models. Used with permission from Taylor et al. [51].

By automating these steps, ARTEMIS streamlines the otherwise laborious and complex task of interface generation, enabling systematic, unbiased exploration of potential heterostructure interfaces . This method has successfully modeled diverse interfaces ranging from semiconductor heterojunctions to 2D/3D material contacts, significantly accelerating computational materials research and enhancing the reliability of interface predictions [51, 52].

When the interface forms between distinct structural phases, such as cubic and hexagonal polymorphs, unique bonding geometries and strain fields arise [11, 17]. For instance, in layered $2D|3D$ systems like $\mathrm{HfS_2|HfO_2}$, the band alignment changes depending on whether the materials are stacked or laterally connected, with lateral interfaces showing increased reconstruction due to bonding mismatch between the different crystalline motifs [55]. These reconstructions are not merely geometric; they result in different electrostatic and electronic behaviours, influencing band offsets and charge transfer dynamics .

Defects, including vacancies, interstitials, dislocations, and grain boundaries, are often intrinsic to interface formation and can profoundly modify interfacial properties [33]. They may serve as scattering centres, recombination sites, or sources of local doping [28]. In grain boundaries of polycrystalline materials, for example, variations in local stoichiometry and strain can create regions of localised metallicity or enhanced dielectric response, as shown in systems exhibiting colossal permittivity .In some systems, such as CCTO ($\mathrm{CaCu_3Ti_4O_{12}}$), interfacial defects and microstructural features have been shown to dominate the macroscopic dielectric properties [52].

In computational studies, predicting the impact of such defects requires careful treatment of their spatial distribution, charge state, and interaction with surrounding lattice atoms Modern approaches like the RAFFLE algorithm allow for ML-driven structure prediction that accounts for void-filling and local bonding environments, enabling more realistic models of defect-laden interfaces [52].

Depending on their nature, interfaces may enhance or impede material performance [9, 28]. This subsection surveys several application domains in which understanding and engineering interfacial properties is essential .

In semiconductor transistor devices, the interface between the gate dielectric and the semiconducting channel is a critical design parameter [45, 59].

As traditional $\mathrm{SiO_2}$ dielectrics are replaced by high-κ materials such as $\mathrm{HfO_2}$ to meet miniaturisation demands, understanding charge transfer and band alignment across interfaces like $\mathrm{HfS_2|HfO_2}$ becomes essential for ensuring device reliability and performance [55]. Notably, in such $2D|3D$ heterostructures, the type and strength of bonding (e.g. van der Waals vs chemical) directly influences charge redistribution and can cause significant deviation from band alignments predicted by Anderson's rule, the vacuum levels of the two materials must be aligned [14].

In layered 2D electronics, especially those using TMDCs, the stacking geometry, layer orientation, and interfacial disorder all strongly influence the band alignment [13]. These factors affect carrier dynamics, optical transitions, and switching behaviour . Traditional models such as Anderson's rule have proven insufficient; corrections using terms such as ΔE_Γ and ΔE_{IF} have been developed to capture hybridisation and interface dipole contributions more accurately [9, 13].

$$\Delta E_\Gamma = \chi_{h,B} - \chi_{h,A} \tag{2.6}$$

$$\Delta E_{\mathrm{IF}} = \alpha \left(\exp\left[\beta \left(\chi_{h,B} - \chi_{h,A} \right) \right] - 1 \right) \tag{2.7}$$

These corrections allow for predictive modelling of heterostructure behaviour and support targeted band offset engineering . Here, $\chi_{h,A}$ and $\chi_{h,B}$ are the hole affinities (ionisation potentials) of materials A and B respectively, and the constants $\alpha = 0.03$ eV and $\beta = 1.91$ Å/eV are empirically fitted from simulation data [13].

Interfaces also govern performance in energy storage systems [28]. In all-solid-state batteries, for example, the interface between electrode and solid electrolyte dictates ion mobility, chemical stability, and space charge layer formation [9, 29]. Such interfacial processes can lead to resistive interphases or dendrite formation, reducing battery life and safety [9, 29]. DFT modelling has shown that interfacial reconstructions and electrostatic gradients can dominate device behaviour [9, 29].

In thermoelectric materials, interfaces are employed to decouple electrical and thermal transport [28]. Interface-induced phonon scattering suppresses thermal conductivity while preserving electronic conductivity; enhancing the thermoelectric figure of merit, ZT [22]. Layered TMDCs and heterostructures with nanostructured interfaces exploit this mechanism to improve thermal management [32].

Across these examples, it is evident that interfaces are not passive boundaries, but active, tunable regions that critically determine macroscopic functionality [17, 28]. Their predictive modelling, through DFT, MLPs, or interface-specific structure search tools like ARTEMIS, remains a central challenge in modern materials design .

2.3 Conventional Prediction Methods

Understanding and predicting the behaviour of material interfaces requires a firm foundation in computational methods that are rooted in quantum and statistical mechanics [51]. This section introduces the core approaches historically used to model interfaces at the atomic and mesoscale level, focusing primarily on first-principles methods and classical simulation techniques . Although each of these methods offers powerful insights, their accuracy, computational cost, and suitability for interface modelling vary substantially .

A primary tool in this field is DFT, a quantum-mechanical framework capable of describing electronic structure with high precision . DFT is well-suited to calculating key interfacial properties such as formation energy, charge redistribution, and band alignment, making it an essential component of most modern interface studies [9]. Within this domain, software packages such as VASP have become widely adopted for their efficient implementations of the Kohn-Sham formalism [11, 29].

However, DFT has limits. Computationally, DFT's cubic scaling with system size makes it computationally prohibitive for large or disordered interface models [19]. As a result, researchers also turn to classical methods, such as Molecular Dynamics (MD) [22, 26]. MD enables the simulation of dynamic processes and thermal fluctuations over nanosecond timescales . It is also adept at capturing thermal effects and atomic rearrangements [26, 47].

These conventional approaches provide the groundwork for predicting critical interfacial properties, including adhesion energy, charge transfer behaviour, and electron or phonon transport . They remain indispensable despite their limitations, particularly when used in tandem . This section surveys these methods in turn, addressing their theoretical underpinnings, practical implementations, and their relative advantages and constraints in the context of interface science .

A foundational step in many interface modelling workflows is lattice matching between two crystalline materials [17, 19]. This involves aligning compatible Miller planes to minimise strain and identify viable supercell geometries . The method introduced by Zur et al. (1984) [63] laid the groundwork for many subsequent algorithms used in tools such as METADISE and ARTEMIS, which systematically enumerate low-strain combinations, slips, terminations, and atomic registries [51] (see Section 2.2). These approaches assume idealised surfaces and often neglect more complex processes such as surface reconstructions, chemical interdiffusion, or local disorder .

DFT remains the most accurate and widely used quantum mechanical method for predicting interfacial properties . It solves the Kohn-Sham equations to determine the electronic ground state, providing access to total energies, charge densities, and band edge positions [20].

MD simulates the atomic trajectories under classical Newtonian mechanics, governed by interatomic potentials . Classical MD is well-suited to studying thermal processes such as diffusion, defect migration, and strain relaxation on picosecond to nanosecond timescales [15].

Density Functional Theory (DFT)

DFT is a first-principles quantum mechanical method used extensively to model the electronic structure of condensed matter systems, including molecules, solids, and, critically, material interfaces [3, 17, 29]. It provides a variational framework in which the ground-state properties of a many-electron system can be obtained from a functional of the electron density $\rho(\mathbf{r})$, rather than from the many-body wavefunction $\Psi(\mathbf{r}_1, \ldots, \mathbf{r}_N)$. This simplification, formalised through the Hohenberg-Kohn theorems, underpins the modern implementation of DFT via the Kohn-Sham approach, where the interacting system is replaced by an auxiliary system of non-interacting electrons subject to an effective potential that captures Coulomb, exchange, and correlation interactions [29].

The theoretical foundation of DFT rests on two key theorems proven by Hohenberg and Kohn in 1964 [24]:

- First Hohenberg-Kohn Theorem: The ground-state properties of a many-electron system are uniquely determined by its electron density $\rho(\mathbf{r})$. This implies that the external potential $V_{\text{ext}}(\mathbf{r})$ (and hence the full Hamiltonian) is a unique functional of $\rho(\mathbf{r})$.

- Second Hohenberg-Kohn Theorem: The true ground-state density minimises the energy functional $E[\rho]$. Therefore, the ground-state energy can be obtained by variationally searching over all physically admissible densities .

These theorems allow the many-body problem to be reformulated in terms of a scalar function $\rho(\mathbf{r})$, enabling a significant reduction in computational complexity compared to solving the full many-electron wavefunction $\Psi(\mathbf{r}_1, \ldots, \mathbf{r}_N)$.

The total Kohn-Sham energy is given by :

$$E_{\mathrm{KS}}[\rho] = T_s[\rho] + \int V_{\mathrm{ext}}(\mathbf{r})\rho(\mathbf{r})\,\mathrm{d}\mathbf{r} + \frac{1}{2}\int\int \frac{\rho(\mathbf{r})\rho(\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|}\,\mathrm{d}\mathbf{r}\,\mathrm{d}\mathbf{r}' + E_{\mathrm{xc}}[\rho], \tag{2.8}$$

where $T_s[\rho]$ is the kinetic energy of the non-interacting electrons, V_{ext} is the external (ionic) potential, and $E_{\mathrm{xc}}[\rho]$ is the exchange-correlation functional, the exact form of which remains unknown .

Each term in the Kohn–Sham energy functional plays a distinct role in approximating the true many-body electronic ground state :

- $T_s[\rho]$ - Non-interacting kinetic energy :

  This term represents the kinetic energy of a fictitious system of non-interacting electrons that yield the same ground-state density as the real, interacting system . It is computed explicitly from the Kohn-Sham orbitals ψ_i :

  $$T_s[\rho] = -\frac{1}{2}\sum_i \int \psi_i^*(\mathbf{r})\nabla^2\psi_i(\mathbf{r})\,\mathrm{d}\mathbf{r}. \tag{2.9}$$

  It is not equal to the true kinetic energy of the interacting system but serves as a tractable approximation .

- $\int V_{\mathrm{ext}}(\mathbf{r})\rho(\mathbf{r})\,\mathrm{d}\mathbf{r}$ - External potential energy :

  This is the Coulomb interaction between the electrons and the nuclei . The potential V_{ext} is usually taken to be the sum of nuclear point charges :

  $$V_{\mathrm{ext}}(\mathbf{r}) = -\sum_I \frac{Z_I}{|\mathbf{r}-\mathbf{R}_I|}, \tag{2.10}$$

  where Z_I and $\mathbf{R}_I$ are the charge and position of nucleus I .

- Hartree energy :
$$E_{\mathrm{H}}[\rho] = \frac{1}{2} \int \int \frac{\rho(\mathbf{r})\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} \, \mathrm{d}\mathbf{r} \, \mathrm{d}\mathbf{r}' \quad (2.11)$$
This is the classical electrostatic (Coulomb) repulsion energy between electron density clouds . It overcounts interaction by including self-interaction, which is ideally cancelled by the exchange-correlation term .

- $E_{\mathrm{xc}}[\rho]$ - Exchange-correlation energy :

  This term captures all many-body quantum effects missing from the previous three terms . It includes :

  - The difference between the true kinetic energy and $T_s[\rho]$.
  - Exchange effects due to the antisymmetry of the wavefunction (Pauli exclusion) .
  - Correlation effects from dynamic electron-electron avoidance .
  -

  It is the only term not known in closed form and must be approximated (e.g. LDA, GGA, hybrid functionals) .

Taken together, this decomposition allows the ground-state energy and density of an interacting electron system to be obtained by solving a system of self-consistent single-particle equations, greatly simplifying the original many-body problem .

Common approximations include the Local Density Approximation (LDA), Generalised Gradient Approximation (GGA), and hybrid functionals [3]. While LDA assumes a locally uniform electron gas and often leads to overbinding, GGA incorporates density gradients and improves accuracy for molecular and inhomogeneous systems [43].

Hybrid functionals, such as HSE06, incorporate a portion of exact exchange and offer improved bandgap and localisation predictions at significantly greater computational cost [43].

DFT has become foundational in materials modelling due to its ability to predict not only total energies and optimised geometries, but also derived properties including electronic band structures, charge densities, dielectric responses, phonon spectra, and interfacial energetics . Its applications to interface modelling, however, are computationally constrained [19, 29].

Physics-based methods, most notably DFT, provide high accuracy for local bonding environments and band structure prediction [1, 29]. However, the computational scaling of standard Kohn-Sham DFT, typically $\mathcal{O}(N^3)$ with respect to system size, renders large or disordered interface simulations intractable without approximation [5].

This $\mathcal{O}(N^3)$ scaling arises from the solution of the Kohn-Sham equations in each self-consistent field (SCF) iteration . The dominant cost comes from constructing and diagonalising the Hamiltonian matrix, whose dimension scales with the number of electrons or basis functions N . The following pseudocode outlines a simplified structure of a typical DFT SCF loop :

Input: Initial electron density ρ_0
Construct Hamiltonian $H[\rho]$ // $\mathcal{O}(N^2)$ matrix assembly
Solve $H[\rho]\psi_i = \epsilon_i\psi_i$ // $\mathcal{O}(N^3)$ diagonalisation
Compute new electron density ρ_{i+1} from occupied states
Check convergence of ρ converged

The $\mathcal{O}(N^2)$ step comes from evaluating pairwise terms in the Hamiltonian, but the diagonalisation step, which extracts eigenstates ψ_i and energies ϵ_i, scales as $\mathcal{O}(N^3)$ using standard linear algebra routines . Since diagonalisation is repeated in each SCF cycle, this cubic scaling is the dominant computational bottleneck in practical DFT workflows .

Even when feasible, DFT struggles to capture extended features such as long-range strain relaxation, misfit dislocations, and spatially diffuse defect states that dominate realistic interface behaviour [17, 31].

All DFT simulations in this project are conducted using the Vienna Ab initio Simulation Package (VASP), a widely used plane-wave-based implementation of Kohn-Sham DFT . VASP employs the projector-augmented wave (PAW) method to model core-valence interactions, and supports a variety of exchange-correlation functionals including GGA-PBE and hybrid HSE06 [29]. Structural relaxations are typically performed with conjugate gradient algorithms until residual forces fall below a chosen threshold (e.g. 0.01eV/Å) [16].

Periodic boundary conditions are employed in all three spatial directions, necessitating the introduction of vacuum regions for slab models and sufficient supercell dimensions to minimise image-image interactions in interface simulations [35].

The simulation input is controlled via the INCAR, POSCAR, KPOINTS, and POTCAR files, with convergence typically assessed by k-point density and energy cutoff tests . Pulay and Broyden mixing schemes are used for charge density convergence, and dipole corrections may be employed for asymmetric slab geometries .

While VASP provides accurate and robust results for interfacial modelling, it remains computationally intensive [9, 29]. For extended or complex interfacial systems, this motivates the introduction of MLPs, which aim to reproduce DFT accuracy [6, 37].

Other Modelling Methods

In addition to DFT, other classes of atomistic modelling, such as MD, has proven valuable for exploring material interfaces, particularly when temporal evolution or thermal effects are of interest [26].

Molecular Dynamics is a time-resolved simulation method in which the trajectories of atoms are computed by numerically integrating Newton's equations of motion [6, 26]. Forces on each atom are derived either from empirical interatomic potentials (classical MD) or quantum-mechanical calculations (ab initio MD, e.g. Car-Parrinello or Born-Oppenheimer approaches) [6]. In interface science, MD is used to study thermal stability, strain relaxation, diffusion, and defect evolution at finite temperatures [26].

One advantage of MD is its capacity to resolve dynamic processes at the atomic level, such as dislocation glide, grain boundary motion, or interfacial melting [11]. However, timescales are severely limited: typical simulations span nanoseconds, far shorter than the seconds to minutes often relevant in experiments . Classical MD can scale to millions of atoms but sacrifices accuracy; ab initio MD is more accurate but typically limited to fewer than 1,000 atoms and short timescales due to computational cost [26].

For interface modelling, MD is particularly useful when studying thermally activated restructuring, interdiffusion, and local vibrational behaviour, especially under conditions such as annealing or irradiation . It also serves as a preliminary stage in workflows involving MLPs, providing dynamical data for training [26].

MD complements DFT in the modelling of interfaces . While DFT offers high-fidelity energetics and electronic structure, MD enables atomistic dynamics over nanoseconds [3, 26, 29]. As such, hybrid approaches are increasingly common, with DFT used to parameterise MD models, or ML techniques introduced to accelerate sampling across scales [9, 26, 47].

What are the advantages and limitations of DFT and MD?

Each computational method traditionally employed in the modelling of interfaces, DFT and MD, offers a distinct balance between accuracy, computational efficiency, and physical scale . While often applied in isolation, their complementary regimes suggest the need for hybrid or hierarchical approaches in the study of real interfaces [29].

DFT is the most widely used ab initio method for predicting atomic-scale properties at interfaces [10]. Crucially, DFT is capable of modelling materials across the periodic table and is the backbone of modern electronic structure prediction . Its limitations include conventional exchange-correlation functionals (e.g. GGA) systematically underestimate band gaps and cannot capture excited-state dynamics without costly extensions such as GW or time-dependent DFT [9, 26].

Molecular Dynamics (MD) offers time-resolved access to thermally activated processes and interfacial dynamics . Classical MD, using empirical or machine-learned interatomic potentials, enables simulation of thousands to millions of atoms over nanosecond to microsecond timescales [6, 26, 47]. However, MD inherits limitations from its force field or potential model: empirical potentials may not transfer well across bonding environments, while high-fidelity MLPs require extensive and chemically diverse training data . Furthermore, standard MD remains constrained to accessible timescales, making it ill-suited for capturing rare events such as defect migration, reconstruction, or long-term phase evolution [26].

How are properties calculated or predicted?

Strain energy quantifies the elastic energy penalty incurred when a material is deformed from its relaxed bulk configuration to match a heterostructure's imposed lattice .

In interface models, this typically arises when one or both materials are forced to adopt a common in-plane lattice constant, leading to tension, compression, or shear .

Assuming the interface uses a uniform lattice derived from the stack, the strain energy for each material can be computed by comparing its relaxed bulk to a strained bulk structure with the same lattice but without the other material present .

Strain Energy

Within DFT, the strain energy E_{strain} is given by :

$$E_{\text{strain}} = \sum_i \left(E_{i,\text{strained}} - E_{i,\text{bulk}}\right), \tag{2.12}$$

where $E_{i,\text{bulk}}$ is the total energy of the relaxed bulk structure of material i, and $E_{i,\text{strained}}$ is the energy of the same structure with lattice vectors deformed to match the interface (but without interface bonding) .

If direct coordinates are used, the atomic basis remains invariant, and all deformation is encoded in the lattice vectors . The sign and magnitude of strain for each material can be determined by comparing the interface lattice a_{stack} with the material's relaxed bulk lattice $a_{i,\text{bulk}}$:

- If $a_{\text{stack}} > a_{i,\text{bulk}}$, the material is under tension .
- If $a_{\text{stack}} < a_{i,\text{bulk}}$, the material is under compression .

This decomposition enables strain energy to be treated as a separate thermodynamic contribution, distinct from interfacial adhesion or chemical reconstruction .

Strain modes in crystalline materials describe the type of mechanical deformation experienced when lattice parameters are altered to accommodate interface formation .

At interfaces, three primary strain types can arise :

- Tensile strain occurs when a material is stretched beyond its equilibrium lattice constant ($\varepsilon > 0$). This is typical for materials with smaller relaxed lattice parameters forced to match a larger interface lattice .
- Compressive strain occurs when a material is squashed to fit a smaller lattice constant ($\varepsilon < 0$) . This affects materials with larger relaxed lattice constants .
- Shear strain involves angular distortions of the unit cell, where interaxial angles deviate from 90, or when off-diagonal components of the strain tensor become nonzero . This may arise from lattice rotation, stepped surfaces, or low-symmetry interface matching .

The total strain state can be expressed by the strain tensor ε, which includes both normal and shear components . For small deformations, the linear strain along a given direction is defined as :

$$\varepsilon = \frac{a_{\text{stack}} - a_{\text{bulk}}}{a_{\text{bulk}}}, \tag{2.13}$$

where a_{stack} is the in-plane lattice constant of the interface, and a_{bulk} is the equilibrium bulk lattice constant of the material .

In complex interfaces, all three strain types may be present simultaneously, and their combined effects influence both the elastic energy and the atomic relaxation pathways .

Surface Energy

Surface energy quantifies the energetic penalty associated with terminating a bulk crystal to expose a surface . This cost arises from the breaking of atomic bonds and the disruption of periodic bonding environments, and is a key thermodynamic quantity in surface science, crystal morphology, and thin film growth .

It is typically computed by comparing the total energy of a slab model with vacuum separation to the energy of an equivalent bulk reference .

Within DFT, the surface energy E_{surf} is defined as :

$$E_{\mathrm{surf}} = \frac{1}{2A}\left(E_{\mathrm{slab}} - n \cdot E_{\mathrm{bulk}}\right), \tag{2.14}$$

where :

- E_{slab} is the total energy of a slab supercell with two exposed surfaces and sufficient vacuum to prevent spurious interactions .
- E_{bulk} is the total energy of the corresponding bulk unit cell (relaxed under identical conditions) .
- n is the number of bulk units contained within the slab supercell .
- A is the surface area of the slab, typically measured in Å^2 .
- The factor of 2 accounts for the two surfaces created in the periodic slab .

The resulting surface energy is reported in units of energy per unit area (e.g. eV/Å^2), and characterises the cleavage cost associated with a particular crystallographic termination .

Surface anisotropy arises because different crystallographic planes expose different bonding environments and coordination numbers, resulting in direction-dependent surface energies . This anisotropy plays a key role in determining equilibrium crystal shapes via the Wulff construction, and strongly affects interface formation when specific facets are matched between materials .

In relaxed slab models, surface reconstruction may occur, further modifying the computed surface energy compared to the ideal cleavage limit . The choice of surface termination, atomic registry, and slab thickness all influence the final value .

Interface Energy

Interface energy quantifies the energetic cost of joining two distinct materials to form a coherent or semi-coherent interface . It reflects the balance between bond formation across the interface and any penalties due to lattice mismatch, structural distortion, or chemical incompatibility .

Within DFT, the interface energy E_{int} is calculated by comparing the total energy of the interface supercell to the energies of the isolated constituent slabs, each relaxed independently but matched in thickness and termination .

The basic expression is :

$$E_{\mathrm{int}} = \frac{1}{2A}\left(E_{AB} - E_A - E_B\right), \tag{2.15}$$

where:

- E_{AB} is the total energy of the full interface supercell .
- E_A and E_B are the energies of the individual slabs A and B (in isolation, matched in structure) .
- A is the cross-sectional area of the interface, as with surface energies typically give in Å^2 .
- The factor of 2 accounts for the two interfaces present in a periodic cell ($A|B$ and $B|A$) .

This formulation captures the net energy change from bringing two surfaces together and allowing relaxation at the junction . It is often used to assess interface stability, compatibility, and bonding quality between dissimilar materials .

However, when constructing interface supercells by lattice matching, strain is often introduced into one or both materials . This strain contributes elastic energy that is not intrinsic to the interface itself . To isolate the true interfacial contribution, a corrected interface energy can be defined by removing the strain components, as described in the previous section :

$$E_{\text{int}}^{\text{corrected}} = E_{\text{int}} - E_{\text{strain},A} - E_{\text{strain},B} \tag{2.16}$$

This correction is especially important in lattice-mismatched systems, where elastic distortion may dominate the raw interface energy . By separating strain and interfacial contributions, this approach yields a more accurate thermodynamic descriptor for interface formation and adhesion .

Formation Energy

Formation energy quantifies the thermodynamic stability of a structure relative to the sum of its isolated, elemental reference phases . It is widely used in materials modelling to assess the favourability of compounds, alloys, and heterostructures, particularly in high-throughput screening and phase stability analysis .

In bulk systems, the formation energy per atom reflects the energy above or below the convex hull of competing phases and is sometimes referred to as the hull energy . A negative formation energy indicates a structure more stable than the linear combination of its elemental constituents, while a positive value implies a metastable or unstable configuration .

In interface models, formation energy provides a normalised metric to compare supercells of varying composition, lattice size, and bonding geometry . It can be computed directly from total DFT energies using one of two equivalent expressions, depending on the number of species involved .

For binary systems containing two species A and B, the formation energy per atom is :

$$E_{\text{form}} = \frac{E_{AB} - \left(n_{A,AB} \cdot \frac{E_{A,\text{bulk}}}{n_{A,\text{bulk}}}\right) - \left(n_{B,AB} \cdot \frac{E_{B,\text{bulk}}}{n_{B,\text{bulk}}}\right)}{n_{AB}}, \tag{2.17}$$

where:

- E_{AB} is the total energy of the interface or alloy supercell .

- $n_{A,AB}$ and $n_{B,AB}$ are the numbers of atoms of each element in the supercell .
- $E_{A,\text{bulk}}$, $E_{B,\text{bulk}}$ are the total energies of the bulk reference structures .
- $n_{A,\text{bulk}}$, $n_{B,\text{bulk}}$ are the number of atoms in the corresponding bulk reference unit cells .
- $n_{AB} = n_{A,AB} + n_{B,AB}$ is the total number of atoms in the structure .

For generalised multi-species systems (e.g. C–Si–Ge–Sn), a compact form is used :

$$E_{\text{form}} = \frac{E_{\text{super}} - \sum_i \left(n_{i,\text{super}} \cdot \frac{E_{i,\text{bulk}}}{n_{i,\text{bulk}}} \right)}{n_{\text{super}}}, \tag{2.18}$$

where the summation runs over all species i present in the supercell . This formulation accommodates arbitrary compositions and is particularly useful when modelling alloyed interfaces or complex heterostructures .

By convention, formation energies are reported in units of energy per atom (e.g. eV/atom) and used to rank relative stabilities across a dataset . However, when comparing structures with different stoichiometries or element counts, normalisation by atom count must be handled carefully to ensure consistent interpretation .

Adhesion Energy

Adhesion energy is a thermodynamic quantity describing the work required to separate two materials at their interface . It is commonly computed from the total energies of the isolated constituents and the relaxed interface configuration . Within DFT, the adhesion energy E_{adh} is given by :

$$E_{\text{adh}} = \frac{1}{A} \left(E_{\text{slab1}} + E_{\text{slab2}} - E_{\text{interface}} \right), \tag{2.19}$$

where A is the interface area, and the energies refer to fully relaxed geometries of the individual slabs and the interface supercell .

Electron transport

Electron transport across interfaces is more nuanced and often requires evaluation of both band alignment and carrier mobility [29, 45]. Band alignment can be predicted via several approaches . The simplest, Anderson's rule aligns vacuum levels to estimate offsets from electron affinities or ionisation potentials [9]. However, this often fails to capture interface-specific effects such as charge transfer, built-in fields, or hybridisation . More accurate predictions therefore rely on self-consistent DFT calculations that extract the local electrostatic potential across the heterostructure to determine band offsets directly [9, 29].

For example, the offset between conduction or valence band edges across an interface can be obtained by aligning the macroscopic average electrostatic potentials of the constituent slabs with that of the full interface . This method accounts for relaxation, dipole formation, and interfacial reconstruction [29]. In cases where electronic coupling is strong, or the valence states are spatially delocalised across layers (as in type-II or type-III band alignment), transport characteristics such as charge separation, carrier confinement, or tunnelling probability can change significantly [32].

2.4 Emergence of ML in Materials Science

Over the past decade, ML has emerged as a transformative tool in materials modelling, offering a route to accelerate predictions and expand accessible system sizes beyond those tractable by first-principles methods [37, 41, 46, 47]. Traditional computational techniques, such as DFT and MD, have established themselves as accurate and rigorous, but they often scale poorly with system size or configurational complexity . This limitation becomes particularly acute in the modelling of interfaces, where subtle geometric, chemical, and electronic variations can dramatically affect material behaviour [6, 29, 37]. ML offers a complementary paradigm, enabling the construction of surrogate models trained on high-fidelity data, which can interpolate or even extrapolate to predict key material properties across vast configuration spaces [6, 43, 46].

This section introduces the growing role of ML in materials prediction, highlighting its application in interface modelling and the broader landscape of data-driven approaches . Subsequent subsections will examine the types of models in use, including classical regressors and deep neural architectures, alongside the nature of training data, typical input descriptors, and validation strategies . Special attention will be given to machine-learned interatomic potentials (MLPs), particularly those with demonstrated success in interfacial contexts such as MACE and ChgNet . Finally, we reflect on the benefits and ongoing limitations of ML methods in materials science, including issues of transferability, uncertainty quantification, and integration into automated modelling pipelines .

How has ML been introduced into materials prediction?

ML has been introduced into materials prediction as a complementary tool to traditional first-principles methods, offering improved scalability, predictive flexibility, and the potential for workflow automation . Its adoption has been driven by limitations in conventional approaches such as DFT, particularly in capturing the structural and chemical complexity of real-world systems such as interfaces [6, 19, 58].

ML enables efficient exploration of vast configurational landscapes, where brute-force DFT sampling would be infeasible . This is particularly important in interface modelling, where multiple stacking orders, terminations, and slip configurations can exist [43]. ML-driven screening allows researchers to rapidly identify promising candidates for further evaluation [6, 19, 46].

Early ML applications focused on surrogate models trained to replicate DFT-derived energies, forces, or properties [6, 15, 46]. Descriptor-based regression models, using quantities such as coordination numbers, Bader charges, or bond lengths, were employed to predict interfacial stability or band alignments . These approaches have evolved toward end-to-end learning frameworks using physically-informed descriptors or graph-based representations [6, 19, 47].

ML is now embedded into iterative prediction–generation–evaluation cycles . For example, predicted interface favourability informs structure generation (e.g. stacking vectors or surface terminations), which are then evaluated by DFT or ML surrogates . The results feed back into model training, supporting a self-refining loop [6, 46, 47]. Tools such as ARTEMIS benefit from such integration [54].

More recent work employs graph neural networks (GNNs), such as MACE or NequIP, which are equivariant to Euclidean transformations and thus well-suited for atomistic systems . These models learn directly from atomic positions and species, avoiding the need for handcrafted features while preserving symmetry constraints essential for interface modelling [6, 15, 37].

Ultimately, ML enables a shift from static modelling pipelines to more autonomous workflows .

What types of models and data are being used?

Models can be broadly categorised into classical learning algorithms and deep learning approaches . Classical methods include random forests, gradient boosting, support vector machines (SVMs), and Gaussian process regression (GPR), the latter being particularly valuable for small datasets due to its uncertainty estimation capability [23]. Deep learning architectures include feedforward neural networks and, more notably, graph neural networks (GNNs), which are well-suited for materials modelling due to their ability to represent variable-sized atomic graphs with complex bonding topologies [6, 15, 43].

The input to these models typically consists of numerical representations of atomic configurations known as descriptors . Geometric descriptors include bond lengths, coordination numbers, angular distributions, n-body distribution functions, and radial symmetry functions; many of which feature prominently in Behler-Parrinello-type neural networks [7,62]. Chemical descriptors such as electronegativity differences and Bader charges capture bonding character and charge transfer [43,46]. Electronic descriptors like partial density of states (PDOS) and local work functions are relevant for predicting band alignment and interfacial transport properties [19]. Interface-specific features, such as interfacial roughness or vacuum level offsets, are particularly crucial in 2D|3D and mixed-dimensional heterostructures [9, 17, 55].

Learning across high-dimensional and heterogeneous structural features requires symmetry-invariant and scale-independent representations, ensuring that physically equivalent systems yield consistent outputs . These enable generalisation across materials with different atom counts or lattice sizes [4, 6, 46]. To support this, models often use learned embeddings; such as message passing networks or autoencoders, to map complex structural data into a lower-dimensional latent space optimised for prediction [4, 6].

The datasets used to train such models vary in size and fidelity . They often derive from high-throughput DFT calculations available in public repositories such as the Materials Project or AFLOW [12, 34].

What are the strengths and limitations of ML in this field?

ML has emerged as a promising strategy for accelerating materials modelling, particularly in the context of complex interfacial systems . Its strengths lie chiefly in efficiency, scalability, and adaptability [6, 26, 47]. ML models, once trained, can rapidly approximate formation energies, surface reactivities, and stability metrics for thousands of configurations, many of which would be prohibitively expensive to assess using first-principles methods alone . This computational efficiency enables exploration of broader configurational and chemical spaces, including systems with disorder, stacking faults, or metastable structures that pose challenges for traditional simulations [6, 43, 47].

Another key advantage is the capacity of ML to capture subtle, non-linear correlations in high-dimensional datasets [6, 47, 57]. When used in conjunction with tools such as ARTEMIS or RAFFLE, ML can assist not only in predicting energetics, but also in prioritising candidate geometries during structure generation [52, 54].

However, ML approaches also exhibit critical limitations . A foremost concern is their dependence on the training dataset: models typically interpolate well within the domain of known examples but generalise poorly to unseen chemistries or extreme geometries . This limits their reliability when extrapolating to novel material classes or interface types, unless training sets are curated with care to ensure coverage of relevant chemical and structural motifs [9, 46, 48].

Another limitation is interpretability . While classical models (e.g. those based on DFT) yield physically grounded outputs, such as charge densities or band structures, many ML models operate as statistical black boxes [6, 46, 48]. Though efforts are underway to integrate physically meaningful descriptors (e.g. n-body distribution functions, local work function, or Bader charges), ensuring transparency and robustness remains a work in progress [6].

Finally, ML alone cannot fully replace physics-based methods . For systems exhibiting emergent interfacial phases, or electronic reconstructions, high-fidelity methods like DFT or many-body perturbation theory remain necessary for validation and refinement . As such, a hybrid strategy is often advocated; using ML to pre-screen candidate structures and flag configurations for further high-accuracy assessment [29, 56, 57].

ML constitutes a valuable addition to the materials modelling toolkit, particularly for screening and guiding interface prediction . Yet its performance remains contingent on data quality and physical context, and it is best deployed in tandem with more rigorous electronic structure methods .

MLPs

MACE

The MACE (Many-body Attention and Correlation Equivariant) model represents a recent advancement in equivariant message passing neural networks (MPNNs), tailored for high-accuracy and scalable force field construction . Unlike earlier MPNNs which rely primarily on two-body invariant messages, MACE employs many-body interactions and equivariant tensor contractions to capture angular and chemical complexity with improved expressivity and data efficiency [5].

One notable strength of MACE is its capacity to reach state-of-the-art accuracy on diverse benchmarks, including rMD17, 3BPA, and AcAc, while requiring only two message passing iterations, due to its use of high-body-order messages . For instance, in extrapolation tasks involving out-of-distribution molecular geometries or high-temperature trajectories, MACE (particularly with $L = 2$) has demonstrated superior performance and robustness relative to competing models such as NequIP and BOTNet [5, 6].

ChgNet

ChgNet is a neural network potential that incorporates both atomic positions and charge distributions into its prediction framework, thereby explicitly learning charge transfer phenomena . This is particularly relevant for interfaces where electrostatics, dipole formation, and local charge accumulation significantly affect electronic properties such as band alignment [15, 40].

Use in Interface Modelling

In the context of this project, MLPs like MACE and ChgNet offer distinct but synergistic advantages . MACE, with its speed, high accuracy, and symmetry-respecting architecture, is well-suited for structure relaxation and force prediction across multi-material junctions [5]. ChgNet, on the other hand, may provide more faithful predictions of electronic interface effects arising from local charge redistribution [15].

Therefore, hybrid workflows that use MLPs for broad structural exploration, followed by targeted DFT recalculations for ambiguous configurations, are increasingly being adopted [19, 43, 46].

MLPs, such as those based on graph neural networks or equivariant message passing architectures, have emerged as promising tools to extend predictive power to larger systems at near-DFT accuracy . Nonetheless, they too face structural limitations . MLPs require extensive, representative training datasets, typically derived from DFT calculations, and their performance deteriorates when applied to configurations involving unseen chemistries, surface reconstructions, or high strain regimes [6, 37, 57]. Furthermore, many ML models lack rigorous uncertainty quantification, making it difficult to detect failure modes in out-of-distribution predictions [28]

Both DFT and ML approaches also fall short in capturing emergent interfacial phenomena such as charge transfer, interface-induced dipoles, or the formation of new interfacial phases . These effects are often strongly dependent on the local atomic registry and chemical heterogeneity, and cannot be inferred from the properties of the isolated constituents alone [9, 37]. Notably, simple models like Anderson's rule have been shown to systematically fail in 2D and 2D|3D heterostructures, necessitating corrections such as ΔE_{Γ} and ΔE_{IF} that explicitly depend on the atomic-scale interface structure and electrostatics [55].

2.5 Limitations of Current Approaches

Despite notable advances in both physics-based and ML approaches to materials modelling, several persistent limitations constrain the predictive fidelity, scalability, and generalisability of current methods, particularly when applied to realistic interface systems [6, 19, 46]. This section outlines three critical domains in which these limitations manifest, motivating the hybrid methodologies explored later in this report:

- What are the current bottlenecks in interface prediction? Interface prediction remains an inherently multiscale problem, combining combinatorial complexity at the atomic level with long-range elastic and electrostatic effects . First-principles approaches such as DFT offer high accuracy but scale cubically with system size, making them infeasible for large or disordered systems typical of interfaces [9, 29]. Even with machine-learned interatomic potentials (MLPs), exploring configurational space, e.g. over terminations, interlayer slips, and misfit reconstructions, requires extensive sampling and often yields flat energy landscapes with many metastable minima [6, 15, 37].

- Where do both ML and physics-based methods fall short? DFT underestimates band gaps and struggles to capture emergent interfacial phenomena such as dipole formation, phase reconstruction, or charge transfer without recourse to hybrid functionals or many-body perturbation theory [9, 29, 43]. Conversely, ML models, especially those trained on bulk or idealised configurations, exhibit limited transferability to chemically complex, strained, or disordered interfaces [57]. Without embedded physical constraints or domain-specific descriptors, ML models often fail to resolve interfacial states, misfit dislocations, or partial intermixing [19, 28].

- Are there data or generalisation issues? Interface-specific datasets are significantly less developed than those for bulk crystals, due in part to the vast configurational diversity of junctions and the lack of standardised repositories . This data sparsity hampers the ability of ML models to generalise across terminations, chemistries, or stacking motifs [9, 29]. Additionally, many early ML workflows lack symmetry-invariant and scale-independent representations, increasing the risk of overfitting to specific atom counts or lattice types [46, 47].

These limitations underscore the need for hybrid workflows that integrate ML potentials (e.g. MACE, ChgNet) for large-scale screening, DFT for high-fidelity validation, and structure-generation tools such as ARTEMIS or RAFFLE for efficient sampling of the configurational landscape . Such frameworks aim to balance accuracy, efficiency, and transferability in modelling the behaviour of realistic interfacial systems [15, 52, 54].

What are the current bottlenecks in interface prediction?

The configurational degrees of freedom, ranging from slip vectors, surface terminations, and intermixing patterns to stacking sequences and misfit accommodation, create a vast structural landscape that resists exhaustive exploration via conventional methods [17, 29, 46].

- Configurational Complexity: At the atomic scale, even between lattice-matched materials, variations in terminations, registry shifts, and interlayer relaxations produce an exponential number of possible configurations . The search space rapidly becomes unmanageable, especially in systems requiring large supercells to capture long-range strain or disorder . This renders brute-force DFT calculations infeasible for comprehensive screening [9, 19].
- Transferability of MLPs: MLPs, such as MACE and ChgNet, offer pathways to accelerate structural screening at near-DFT accuracy [5, 15]. However, their performance is contingent upon the representativeness of the training data .

Where do both ML and physics-based methods fall short?

Despite significant progress in both first-principles and ML approaches, key limitations persist in the predictive modelling of material interfaces . These limitations arise not only from computational constraints but also from intrinsic mismatches between real interfacial complexity and the assumptions underpinning current models [19, 56].

Physics-based methods, most notably DFT, provide high accuracy for local bonding environments and band structure prediction . Even when feasible, DFT struggles to capture extended features such as long-range strain relaxation, misfit dislocations, and spatially diffuse defect states that dominate realistic interface behaviour [6, 19].

MLPs require extensive, representative training datasets, typically derived from DFT calculations, and their performance deteriorates when applied to configurations involving unseen chemistries, surface reconstructions, or high strain regimes [6, 37]. Furthermore, many ML models lack rigorous uncertainty quantification, making it difficult to detect failure modes in out-of-distribution predictions [6, 15].

Are there data or generalisation issues?

In contrast to bulk crystals, where structured repositories like the Materials Project or AFLOW provide extensive coverage, interface-specific datasets remain sparse . This is due to the combinatorial explosion of viable terminations, slip vectors, and orientations across material pairs, especially in mixed-dimensional systems [15, 19, 28]. Existing databases tend to overrepresent a narrow set of lattice-matched, technologically prominent interfaces, introducing bias and hindering the model's capacity to generalise across broader chemical or structural classes [13, 17, 19].

Another critical challenge is the lack of standardisation in how interface data is represented . Unlike bulk materials, which can be described relatively easily by a set of structural features such as lattice constants and atomic compositions, interfaces require more complex descriptors, including local bond environments, atomic coordination numbers, and structural motifs that change drastically depending on orientation, termination, and strain [19, 57]. As a result, the design of robust and transferable ML models for interfaces is further complicated by the difficulty in developing universally applicable features that capture the intricate nature of interfacial systems [26, 56]. This problem is compounded by the lack of large, consistent datasets that integrate structural, energetic, and electronic data, making it harder for models to learn the subtle interactions that govern interface stability and properties .

Finally, the sheer diversity of possible interface configurations, especially for systems involving two-dimensional (2D) materials, complex oxides, or disordered solids, makes high-throughput DFT calculations or other first-principles methods impractical for exhaustive data collection [11, 17]. To address these issues, there is a pressing need for hybrid data collection strategies that combine high-fidelity, physics-based methods with more scalable, data-driven approaches, such as MLP models that can rapidly generate large datasets of interfacial configurations [6, 19, 48]. Additionally, targeted experimental datasets that validate theoretical predictions are essential for creating reliable and comprehensive interfaces databases [19, 22, 46].

Generalisation error is further amplified by suboptimal representation of interface-specific physics . Many descriptor schemes, such as SOAP vectors or radial symmetry functions, are optimised for bulk periodicity and fail to capture interfacial asymmetry, vacuum discontinuities, or broken symmetry [62]. Representations that do not enforce scale-invariance or rotational equivariance can also hinder model transferability to supercells or faceted configurations [6, 56].

Finally, inconsistencies in training labels present an often-overlooked source of predictive error . Energies obtained from high-throughput DFT pipelines can vary depending on pseudopotential choice, relaxation tolerances, or convergence thresholds [43, 56]. In interfacial contexts, these uncertainties are magnified by metastability and complex relaxation pathways. .

Some datasets rely on static, unrelaxed geometries or surrogate models, further undermining the reliability of ground truth energetics [19, 56].

While ML offers substantial speed-ups over DFT, its utility in interface prediction is presently constrained by data bottlenecks, representational mismatch, and transferability failures . Addressing these limitations will require a multi-pronged strategy involving broader and more diverse training datasets, improved descriptors attuned to interfacial physics, and integration of uncertainty quantification into ML workflows . Hybrid schemes, where MLPs perform large-scale screening followed by selective DFT refinement, may offer a viable path forward .

2.6 State-of-the-Art in Interface Prediction

Recent years have seen rapid developments in the modelling of interface structures, driven both by methodological advances and by the growing technological importance of heterostructured materials [9, 19, 26]. Interface prediction now spans a range of computational approaches, from first-principles calculations and high-throughput screening to data-driven and ML methods [26, 41]. This section provides an overview of the current state-of-the-art, establishing the context for the subsequent discussion of recent studies (Section 2.5.1), the positioning of this project's methodology (Section 2.5.2), and anticipated emerging directions (Section 2.5.3).

A key enabler of recent progress has been the development of automated interface generation tools such as ARTEMIS and InterMatch, which systematically enumerate possible lattice matches, terminations, and stacking configurations between two crystal structures . When integrated with rapid energy evaluation, either via DFT or MLPs, these tools allow the configurational space of interface structures to be sampled more broadly and with greater fidelity [19, 51]. In parallel, studies have increasingly sought to move beyond idealised, coherent interfaces to include semi-coherent, faceted, and amorphous boundaries, reflecting the complexity of real materials under fabrication conditions [13, 26, 38].

Efforts have also been made to improve the prediction of interfacial electronic properties, such as band alignment . Where traditional rules like Anderson's rule fail, corrective models based on physically grounded terms, such as ΔE_{Γ} and ΔE_{IF}, have been proposed and validated across a range of 2D materials [13]. Such corrections are essential for capturing charge transfer and hybridisation effects at interfaces, and form an important bridge between local structure and electronic response .

This chapter aims to synthesise these various advances, situating the present work within an evolving landscape of interface modelling . Particular attention will be given to scalable workflows that combine structural generation, energetic evaluation, and property prediction, setting the stage for a more autonomous and predictive framework for materials interface discovery .

Hybrid methodologies now dominate the landscape of ab initio interface prediction . Notable among them is the ARTEMIS tool developed by Taylor et al., which automates the generation of candidate interface structures by identifying lattice-matched slabs, multiple surface terminations, and lateral shifts across a wide range of Miller planes [51].

Shortly after, Taylor et al. introduced the RAFFLE, with Pitfield et al. introducing methodology, combining empirical void-filling algorithms with iterative statistical descriptors to model metastable and low-symmetry reconstructions [52].

Constrained variants of the Minima-Hopping Method (MHM) allow for controlled exploration of grain boundary reconstructions by varying atomic densities and introducing geometric constraints [50]. These approaches yield more realistic reconstructions than fixed-stoichiometry DFT or classical potentials, especially in systems like silicon where interface states are sensitive to atomic arrangement and misfit accommodation .

ML is increasingly embedded into the prediction workflow. Gerber et al. developed the InterMatch framework, which leverages high-throughput DFT databases to train ML models that guide lattice matching and stacking predictions in 2D systems [19]. At the electronic level, Davies et al. proposed physically motivated corrections to band alignment, namely ΔE_{Γ} and ΔE_{IF}, to extend Anderson's rule and account for hybridisation and interface dipoles in 2D heterostructures [13]

these advances, several limitations persist . Data scarcity for interfacial systems hampers generalisation, and most ML models are trained on bulk-like environments . Existing descriptors may fail to capture the broken symmetry, local dipoles, and charge redistribution unique to interfaces . Furthermore, many workflows truncate the configurational space prematurely or rely on geometric heuristics alone [19, 26, 41].

There is increasing recognition that hybrid workflows, combining ML surrogates with tools like ARTEMIS and RAFFLE, and validating critical configurations using DFT, offer a scalable route forward . This multiresolution strategy enables both efficient exploration and physically grounded evaluation of interfacial stability across diverse systems [51, 52].

How does my proposed approach compare and build upon it?

Contemporary approaches to interface structure prediction combine symmetry-based lattice matching, high-throughput screening, and DFT relaxation . Tools such as ARTEMIS enable the systematic generation of interface candidates by identifying lattice commensurabilities and constructing multiple terminations and stacking arrangements from parent surfaces [54]. However, the energetic evaluation of these candidates remains a bottleneck: full relaxation and total energy comparison using DFT is prohibitively expensive for systems exceeding a few hundred atoms .

To mitigate this, recent advances have introduced surrogate strategies that bypass exhaustive DFT sampling . InterMatch, a high-throughput framework proposed by Gerber et al., guides stacking predictions using pre-computed DFT databases and electrostatic heuristics [19]. Yet such frameworks often lack spatial resolution or generalisability across chemically diverse or defective systems [19, 26].

Emerging trends

Recent advances in interface prediction have increasingly centred on integrating data-driven approaches with atomistic modelling frameworks . MLPs such as MACE and E(3)-equivariant networks like NequIP are now enabling the efficient evaluation of interfacial configurations across large configurational spaces . This shift permits relaxation of structures at near-DFT accuracy, allowing researchers to explore interface separations, registry shifts, and metastable motifs that would be prohibitively expensive using standard ab initio methods [5, 6, 37].

Collectively, these approaches suggest a field increasingly focused on scalable, automatable, and physically interpretable modelling of interfaces . The combination of ML with established electronic structure theory is opening a path toward predictive frameworks that are sensitive to real-world synthesis constraints and device contexts .

Chapter 3

Investigation of Interface Favourability

A key step in training a MLI model is building a dataset of physically plausible and energetically reasonable heterostructures [6, 17]. Ideally, such a dataset would be generated entirely using DFT, but the cost of relaxing and evaluating thousands of configurations makes this impractical [43, 43, 47]. Instead, this chapter investigates whether an MLP, specifically the MACE model, can assist in rapidly generating and assessing interface structures before applying DFT.

The focus here is not to validate MLPs as a replacement for DFT, but to test their usefulness as part of a larger data-generation pipeline. If MACE is able to help identify realistic and favourable structures, it can reduce the number of expensive DFT calculations needed and enable faster construction of training data for the MLI model.

The chapter is structured as follows: Section 1 describes the generation of the dataset, the chemical systems considered, and the use of ARTEMIS for constructing candidate interfaces. Section 2 analyses how well the MACE model reproduces trends in DFT-calculated interface energetics. Section 3 discusses and summarises analysis and findings.

3.1 Methodology

The goal of this study is to support the development of an MLI model that can predict which heterostructures are likely to be realistic and favourable. To train this model, we need a large dataset of physical interfaces that are not only valid in terms of their atomic arrangement but also low enough in energy to be found in real materials. Relaxing and evaluating all possible interfaces using DFT would be too expensive, so we explore whether a Machine-Learned Potential (MLP), specifically MACE, can help us identify good candidates more efficiently.

Rather than replacing DFT, MACE is tested here as a screening tool: if MACE can roughly preserve the order of interface favourability that DFT gives, it could help us pick which interfaces to relax with DFT. These comparisons are a means to an end, providing the MLI model with useful training data.

This work uses an iterative approach to building the dataset. At each step, the number of materials, structures, and configurations is increased to explore broader behaviours and to identify issues in the workflow before scaling further. Each iteration reflects what was possible with available computing resources and analysis at the time.

- Iteration 1 focused on a small number of $Si|Ge$ interfaces generated and relaxed manually to test early code and analysis scripts.
- Iteration 2 expanded this to a much larger $Si|Ge$ dataset using automated tools like ARTEMIS, introducing slab shifting and filtering steps.
- Iteration 3, which forms the main dataset used in this chapter, includes materials from group 14: carbon (C), silicon (Si), germanium (Ge), and tin (Sn).

These elements were chosen because they are widely used in electronics and all have four valence electrons, which avoids problems with charge imbalance when making interfaces. Elements heavier than atomic number 83 are excluded to avoid needing relativistic corrections, it is also notable that MACE is trained on elements $Z \leq 89$.

This third dataset uses a consistent generation workflow based on Miller plane selection, surface matching, and stacking with registry-aware shifts. It avoids configurations with overlapping atoms and stores each system in a reproducible format for later energy evaluation.

3.1.1 Interface Generation Protocol

Interfaces were generated using the ARTEMIS toolkit (see Section 2.2 for background on interface construction and lattice matching). For each pair of materials, up to 64 surface orientations (Miller planes) were explored to find possible ways the two materials could be stacked together. ARTEMIS searched for matching supercells by expanding the unit cells of both materials up to $8\times$ along the in-plane directions, looking for configurations with similar lattice vectors and angles (detailed in Section 2.2). This helps distribute mismatch over more atoms and reduce the per-cell strain [51].

A key goal in finding physically favourable interfaces is to minimise the energy penalties introduced by strain and unbonded valence electrons [9,19]. Interfaces with large lattice mismatch or poor atomic registry often exhibit elevated formation energies due to elastic distortions or chemically unstable bonding environments [11,22,60]. By contrast, coherent interfaces that exhibit low strain and high atomic registry are more likely to be energetically stable and experimentally realisable [11,13,19].

One strategy for achieving this is through supercell scaling [51]: we try to match lattice constants a and a' of two materials by finding integers n and m such that $n \cdot a \approx m \cdot a'$, distributing the residual mismatch over multiple unit cells (see discussion of supercell mismatch reduction in Section 2.2). For example, silicon and germanium have lattice constants of 5.45 Å and 5.66 Å, respectively. Matching 22 unit cells of Si to 21 of Ge gives:

$$|22 \cdot 5.45 - 21 \cdot 5.66| = 1.04 \text{ Å} \tag{3.1}$$

which corresponds to a mismatch of approximately 0.047 Å per unit cell. Scaling up to $2201 Si$ and $2119 Ge$ cells reduces this to:

$$|2201 \cdot 5.45 - 2119 \cdot 5.66| = 1.91 \text{ Å} \quad \Rightarrow \quad \sim 0.0009 \text{ Å mismatch per cell} \tag{3.2}$$

Each increase in supercell size decreases the mismatch per unit cell by roughly an order of magnitude.

In addition to supercell scaling, the choice of interface orientation, described by Miller indices (explained in Section 2.2), can significantly influence atomic registry across the interface. Each Miller plane represents a unique slice through the crystal lattice that exposes a specific atomic arrangement [51]. By rotating, stacking, and aligning these planes between two different materials, one can search for surface combinations where atoms naturally align into coherent patterns. ARTEMIS uses these surface alignments to explore many candidate configurations, seeking high registry where bonding across the interface is maximised and valence electrons are not left unpaired [51].

Each matched interface was constructed from two slabs, each 6 atomic layers thick, separated by a small vacuum gap. ARTEMIS then applied systematic shifts in the a and b directions to explore stacking offsets between the slabs (details of which are discussed in Section 2.2). Initially, zero-shift configurations were allowed, but these sometimes produced unphysical atomic overlaps along the c-axis. These cases were filtered out during data cleaning. Future datasets will attempt to reintroduce shift-zero configurations in a and b while borrowing c-axis separations from nearby stable interfaces.

Although ARTEMIS supports atomic swaps at the interface to explore alloy-like or disordered bonding environments [51], this feature was not available in the version used for this study. Such configurations will be generated in future iterations using the RAFFLE tool.

While ARTEMIS considered many potential surface combinations, only 29 unique surfaces were ultimately included in the final dataset. This reflects both the geometric constraints of lattice matching and the filtering applied to ensure physical plausibility of the generated structures.

All parts of the workflow, from building interfaces to reading the results, were automated using Python scripts. This made it easier to run large numbers of calculations and keep track of everything.

Interfaces were generated and prepared for calculation using scripts that could run in bulk. Each job was submitted to the compute cluster automatically.

Some calculations failed or gave broken outputs (like NaN energies). These were filtered out automatically so they didn't affect the results.

The energy values from both MLP and DFT were read and stored in CSV files. These files were used to calculate rankings and compare the methods.

3.1.2 Ranking and Error Metrics

To check how well the MLP results agree with DFT, we used a few simple measures.

Spearman ρ

This checks if the overall order of interface energies is preserved between methods. It compares the rankings and gives a score between -1 (completely reversed) and 1 (perfect match):

$$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)} \tag{3.3}$$

Here d_i is the difference in rank for the i^{th} structure, and n is the total number of structures.

Kendall τ

This checks whether pairs of structures are ranked in the same order between methods. It also ranges from -1 to 1:

$$\tau = \frac{\text{(concordant pairs)} - \text{(discordant pairs)}}{\frac{1}{2}n(n-1)} \tag{3.4}$$

Top-N Overlap

his looks at how many of the best-ranked structures under DFT also appear in the top group under MLP. For example, how many of the top 5 or top 10 are shared.

Mean Absolute Error (MAE)

This shows the average difference in energy between MLP and DFT:

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n} |y_i - \hat{y}_i| \tag{3.5}$$

Root Mean Square Error (RMSE)

This is another way of showing how different the energies are, giving more weight to large differences:

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{3.6}$$

These metrics help to determine whether the MLP is giving similar energy rankings and values to DFT, so it can be established whether it is useful for selecting structures for further study.

3.1.3 Relaxation and Energy Evaluation

In this study, the interface structures were not relaxed before comparing energies. Instead, the same unrelaxed geometries were evaluated using both DFT and MLP to keep the comparison fair. This helps isolate differences in energy prediction from differences caused by structural relaxation [51].

DFT calculations were carried out using the VASP software. Gamma-point sampling was used for simplicity and speed. Calculations were limited to a 24-hour walltime. Structures that failed to complete were excluded from analysis.

The MACE model used here was the most recent release available at the time (downloaded in April 2025). It was applied using the ASE library to compute total energies on the same unrelaxed structures as used for DFT. No fine-tuning or delta-correction was applied.

While MLP-based relaxation was tested on some structures later in the workflow, those results were not used for ranking comparison in this section. The goal is to test how well MLP can predict relative stability based on geometry alone. DFT relaxations are planned for future datasets.

3.2 Results and Analysis

This section presents the results of comparing unrelaxed DFT and MLP (MACE) energy rankings across 680 valid heterostructures, grain boundaries, and alloyed systems involving the group 14 elements (C, Si, Ge, Sn). The goal is to determine whether MACE, a machine-learned interatomic potential, can replicate DFT-derived rankings on unrelaxed geometries closely enough to serve as a pre-screening tool in the MLI data pipeline.

Across all evaluated systems, Spearman's rank correlation coefficient was extremely high:

$$\rho = 0.9954, \qquad \tau = 0.9568 \tag{3.7}$$

This near-perfect alignment demonstrates that MACE preserves the order of interface favourability remarkably well, even without relaxation, suggesting strong potential as a filtering method for DFT candidate selection.

However, absolute energy agreement was poor:

$$\text{RMSE} = 188.89 \text{ eV}, \qquad \text{MAE} = 30.87 \text{ eV} \tag{3.8}$$

These large discrepancies reflect the fact that while MACE captures relative stability, its raw energy outputs are not quantitatively reliable without further training. Despite this, the Top-10 overlap between DFT and MLP was $8/10$, indicating strong alignment in identifying favourable structures.

To understand how interface geometry impacts ranking fidelity, results were grouped into three categories: upper-slab heterostructures, lower-slab heterostructures, and homostructures (grain boundaries). Average metrics across these categories are summarised in Table 3.1.

- Upper interfaces: MACE achieved the best performance here with $\rho \approx 0.98$ and Top-10 overlap of $8.5/10$.

  Although RMSE remained high ($\approx 120eV$), the rankings were highly reliable.

- Lower interfaces: Rank correlation dropped to $\rho \approx 0.62$, suggesting geometry-related artefacts when certain species (e.g. carbon) appeared in the lower layer.

- Grain boundaries: Strong correlation was retained ($\rho \approx 0.94$, $\tau \approx 0.83$), and energy error was much lower (RMSE $\approx 7.69eV$), making this group especially reliable.

Carbon Systems $(C|X)$

Only three $C|Si$ interfaces were successfully processed with carbon in the lower layer. These yielded poor results, including a negative rank correlation ($\rho = -0.5$), high error (MAE ≈ 21.6 eV), and a Top-10 overlap of just 3. These structures are excluded from further analysis.

In contrast, when carbon occupied the upper slab (e.g. $C|Si$), the model performed extremely well:

$$\rho = 0.9491, \qquad \tau = 0.8182, \qquad \text{MAE} \approx 0.58 \text{ eV}, \qquad \text{Top-10 overlap} = 8 \tag{3.9}$$

This suggests that MACE can accurately rank and evaluate carbon-containing interfaces when the geometry is physically sensible, particularly when carbon forms the terminating layer rather than the substrate.

Silicon Systems $(Si|X)$

Silicon-rich systems exhibited strongly asymmetric performance depending on whether Si occupied the lower or upper slab.

Table 3.1: Average metrics for each interface type.

Interface Type	N	Spearman ρ	Kendall τ	RMSE (eV)	MAE (eV)	Top-10 Overlap
C|X or X|C	82	0.954	0.830	4.484	1.346	8
C|X	3	-0.500	-0.333	21.575	21.575	3
C|Si	3	-0.500	-0.333	21.575	21.575	3
X|C	79	0.949	0.818	1.787	0.578	8
Si|C	79	0.949	0.818	1.787	0.578	8
Si|X or X|Si	450	0.990	0.941	232.067	45.848	8
Si|X	350	0.991	0.947	238.896	45.297	8
Si|C	79	0.949	0.818	1.787	0.578	8
Si|Ge	150	0.908	0.836	364.914	104.743	8
Si|Sn	105	0.988	0.920	1.807	0.748	8
X|Si	100	0.989	0.931	206.397	47.779	10
C|Si	3	-0.500	-0.333	21.575	21.575	3
Ge|Si	58	0.966	0.900	170.898	40.365	10
Sn|Si	23	0.978	0.897	333.921	102.330	10
Si|Si	16	0.956	0.846	1.251	1.147	10
Ge|X or X|Ge	506	0.990	0.936	207.183	36.806	4
Ge|X	229	0.994	0.952	86.663	11.583	10
Ge|Si	58	0.966	0.900	170.898	40.365	10
Ge|Sn	104	0.991	0.952	1.539	0.828	9
X|Ge	277	0.982	0.927	268.705	57.659	8
Si|Ge	150	0.908	0.836	364.914	104.743	8
Sn|Ge	60	0.977	0.911	0.904	0.582	5
Ge|Ge	67	0.982	0.929	19.584	3.361	10
Sn|X or X|Sn	322	0.996	0.958	89.258	8.130	8
Sn|X	98	0.990	0.936	161.773	24.702	10
Sn|Si	23	0.978	0.897	333.921	102.330	10
Sn|Ge	60	0.977	0.911	0.904	0.582	5
X|Sn	224	0.997	0.962	1.722	0.879	8
Si|Sn	105	0.988	0.920	1.807	0.748	8
Ge|Sn	104	0.991	0.952	1.539	0.828	9
Sn|Sn	15	0.878	0.725	2.241	2.154	9

System	n	ρ	τ	MAE (eV)	RMSE (eV)	Top-10
C_lower_or_upper	82	0.95	0.83	1.35	4.48	8
C_lower	3	-0.50	-0.33	21.58	21.58	3
C_lower_Si_upper	3	-0.50	-0.33	21.58	21.58	3
C_upper	79	0.95	0.82	0.58	1.79	8
C_upper_Si_lower	79	0.95	0.82	0.58	1.79	8

Table 3.2: Performance of MACE on carbon-containing interface systems.

System	n	ρ	τ	MAE (eV)	RMSE (eV)	Top-10
Si_lower_or_upper	450	0.99	0.94	45.85	232.07	8
Si_lower	350	0.99	0.95	45.30	238.90	8
Si_lower_C_upper	79	0.95	0.82	0.58	1.79	8
Si_lower_Ge_upper	150	0.91	0.84	104.74	364.91	8
Si_lower_Sn_upper	105	0.99	0.92	0.75	1.81	8
Si_upper	100	0.99	0.93	47.78	206.40	10
Si_upper_C_lower	3	-0.50	-0.33	21.58	21.58	3
Si_upper_Ge_lower	58	0.97	0.90	40.37	170.90	10
Si_upper_Sn_lower	23	0.98	0.90	102.33	333.92	10
Si_lower_and_upper	16	0.96	0.85	1.15	1.25	10

Table 3.3: Performance of MACE on silicon-containing interface systems.

When silicon was in the lower slab, such as in $Si|Ge$ interfaces, rank correlation remained high:

$$\rho = 0.9082, \qquad \tau = 0.8360, \qquad \text{MAE} = 104.74 \text{ eV} \tag{3.10}$$

Despite good rank alignment, the absolute energy agreement was extremely poor-highlighting significant error in MLP relaxation or energy evaluation for these systems. In contrast, the reversed configuration with silicon in the upper slab ($Ge|Si$) yielded both higher correlation and markedly lower error:

$$\rho = 0.9664, \qquad \tau = 0.8996, \qquad \text{MAE} = 40.37 \text{ eV} \tag{3.11}$$

This asymmetry suggests that silicon's role in anchoring the interfacial bonding network may contribute to model breakdown, especially when acting as a strained or undercoordinated substrate. The MACE model likely struggles to generalise across such distortions.

Germanium Systems

Germanium interfaces showed weaker overall performance compared to other group IV elements:

System	n	ρ	τ	MAE (eV)	RMSE (eV)	Top-10
Ge_lower_or_upper	506	0.99	0.94	36.81	207.18	4
Ge_lower	229	0.99	0.95	11.58	86.66	10
Ge_lower_Si_upper	58	0.97	0.90	40.37	170.90	10
Ge_lower_Sn_upper	104	0.99	0.95	0.83	1.54	9
Ge_upper	277	0.98	0.93	57.66	268.70	8
Ge_upper_Si_lower	150	0.91	0.84	104.74	364.91	8
Ge_upper_Sn_lower	60	0.98	0.91	0.58	0.90	5
Ge_lower_and_upper	67	0.98	0.93	3.36	19.58	10

Table 3.4: Performance of MACE on germanium-containing interface systems.

$$\rho = 0.7360, \qquad \tau = 0.6387, \qquad \text{MAE} = 3.26 \text{ eV}, \qquad \text{Top-10 overlap} = 3 \tag{3.12}$$

Despite moderate rank preservation, the low Top-10 overlap and relatively high error suggest inconsistencies in MLP predictions for Ge-based systems. This may reflect greater structural variability or limited training representation for germanium-containing slabs. Nonetheless, no catastrophic inversions were observed, and the rank correlation is still usable in large-scale screening contexts.

Tin Systems Sn systems performed significantly better across all metrics:

$$\rho = 0.8792, \qquad \tau = 0.7451, \qquad \text{MAE} = 2.06 \text{ eV}, \qquad \text{Top-10 overlap} = 8 \tag{3.13}$$

System	n	ρ	τ	MAE (eV)	RMSE (eV)	Top-10
Sn_lower_or_upper	322	1.00	0.96	8.13	89.26	8
Sn_lower	98	0.99	0.94	24.70	161.77	10
Sn_lower_Si_upper	23	0.98	0.90	102.33	333.92	10
Sn_lower_Ge_upper	60	0.98	0.91	0.58	0.90	5
Sn_upper	224	1.00	0.96	0.88	1.72	8
Sn_upper_Si_lower	105	0.99	0.92	0.75	1.81	8
Sn_upper_Ge_lower	104	0.99	0.95	0.83	1.54	9
Sn_lower_and_upper	15	0.88	0.73	2.15	2.24	9

Table 3.5: Performance of MACE on tin-containing interface systems.

These results indicate strong rank preservation and moderate energy agreement. Tin-containing interfaces appear well-represented in the MACE training set and are consistently evaluated with low error, especially in top-ranked configurations. MLP predictions are therefore considered reliable for Sn-based systems across most tested geometries.

Grain Boundaries (Homostructures)

Grain boundary results were strong overall but showed species-specific variation. Rank correlation was consistently high across all homostructures, though errors varied by material:

- $Ge|Ge$: $\rho = 0.9818$, $\tau = 0.9291$, MAE ≈ 3.36 eV
- $Si|Si$: $\rho = 0.9557$, $\tau = 0.8462$, MAE ≈ 1.15 eV
- $Sn|Sn$: $\rho = 0.8779$, $\tau = 0.7255$, MAE ≈ 2.15 eV

Grain boundaries also achieved the highest Top-10 overlap ($mean = 9.67/10$) and substantially lower RMSE ($\approx$ 7.7 eV) compared to heterostructures. This likely reflects their structural symmetry and stoichiometric balance, which reduce both configurational variability and extrapolation error.

n	ρ (Spearman)	τ (Kendall)	RMSE (eV)	MAE (eV)	Top-10 Overlap
16	0.956	0.846	1.251	1.147	10
67	0.982	0.929	19.584	3.361	10
15	0.878	0.726	2.241	2.154	9

Table 3.6: Summary of MLP vs DFT performance for homostructure grain boundaries.

Perfect rank preservation

$Si|Ge$ alloy systems exhibited near-perfect rank alignment between DFT and MLP evaluations . Both ρ and τ approached 1.0 across the board, as seen in the table below :

Table 3.7: Comparison of unrelaxed formation energies between DFT and MLP for perfect alloys. Despite large absolute energy errors, rank correlation is perfect.

Spearman ρ	Kendall τ	RMSE (eV)	MAE (eV)	Top-6 Overlap
1.00	1.00	3.87	3.86	6/6 (1.0)

Potential Reason

Many systems in this study exhibit chemically uniform environments with coherent, low-strain interfaces. These characteristics likely ensure that atomic configurations remain within the interpolation regime of the MACE model, resulting in stable rank preservation.

MLP evaluations provided orders-of-magnitude speedups over DFT while retaining a strong rank correlation in well-behaved systems. However, this performance is highly system-dependent. Interfaces with silicon in the lower slab-especially those involving strain or geometric asymmetry, showed good rank correlation but extremely poor energy agreement. This suggests that while MACE can correctly rank configurations in some silicon-rich systems, it struggles to capture their absolute energetics when surface termination or bonding environments fall outside the training domain.

These findings support the use of MACE as a high-throughput prefilter for ranking heterostructures prior to DFT refinement. Although absolute energy predictions remain unreliable in some asymmetric or strained systems, particularly when silicon occupies the lower slab, the model consistently preserves rank in chemically ordered, well-coordinated interfaces.

These observations highlight the importance of geometric configuration in model reliability. Rather than a general failure in Si-rich systems, the errors appear linked to bonding environment and slab orientation, motivating future efforts to refine model generalisation through delta-learning or targeted retraining (Chapter 4).

3.3 Summary of Findings

This study evaluated whether machine-learned interatomic potentials (MLPs), specifically the MACE framework, can replicate DFT-derived rankings of interface stability across group-IV systems, including carbon (C), silicon (Si), germanium (Ge), and tin (Sn). The investigation involved automated structure generation via ARTEMIS, energy evaluation using both DFT and MLP on unrelaxed geometries, and statistical analysis of ranking agreement using Spearman ρ, Kendall τ, and Top-N overlap.

Rather than comparing relaxed geometries, all interfaces were evaluated in their unrelaxed form to isolate differences in energy prediction. This enabled a fair, geometry-matched comparison of MLP vs DFT across a large dataset, while identifying how system composition and stacking geometry impact rank preservation.

Agreement between MLP and DFT was found to be strongly system-dependent. MLPs performed exceptionally well for Ge and Sn systems, achieving high rank correlation ($\rho > 0.97$) and Top-10 overlap ($\sim 8/10$) with DFT rankings. For Si-based systems, especially those where Si appeared in the lower slab, energy agreement deteriorated significantly (e.g. MAE > 100 eV), despite high rank correlation ($\rho \approx 0.91$), suggesting sensitivity to stacking order and geometric artefacts. These discrepancies appeared systematic rather than statistical, indicating that MLPs may struggle to accurately predict absolute energies in strained or under-coordinated configurations, even when rank preservation is maintained.

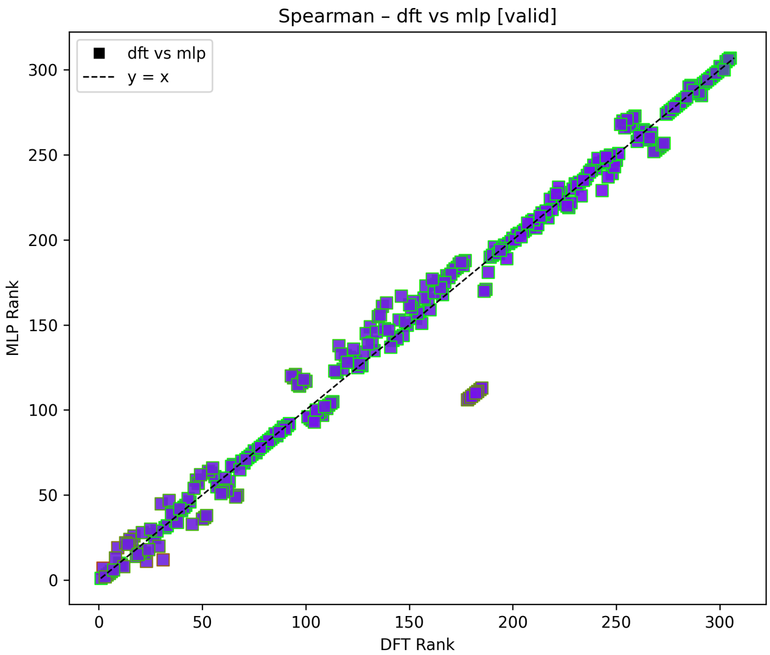

Figure 3.1: Spearman rank correlation for Si interfaces, showing strong rank preservation even in lower-Si systems, despite energy disagreement due to geometric artefacts and interfacial strain.

Carbon results were geometry-dependent: only three valid $C|Si$ structures with carbon in the lower slab were obtained, and these yielded unreliable results ($\rho = -0.5$), leading to their exclusion from analysis. However, when carbon appeared in the upper slab ($C|X$), rank correlation was excellent ($\rho = 0.95$), and MAE dropped below 1 eV. Silicon systems, particularly Si|Ge, showed strong rank correlation but extremely poor energy agreement, while $Ge|Si$ demonstrated improved alignment. Germanium and tin systems showed robust rank preservation and strong Top-N screening performance across the board.

Although direct MLP-on-DFT-relaxed structure evaluations (DFT-relaxed MLP) were not included due to computational cost, the study revealed that relaxation and evaluation consistency (e.g. full MLP vs full DFT) influence error, and future cross-evaluation may further clarify error sources.

Table 3.8: Performance metrics for Si interfaces (Si in upper slab). Lower-slab Si configurations exhibited large energy errors but still maintained reasonable rank correlation (not shown), highlighting geometry sensitivity.

Comparison	Spearman	Kendall	RMSE (eV)	MAE (eV)	Top-10 Overlap
X|Si 100	0.989	0.931	206.397	47.779	10
C|Si 3	-0.500	-0.333	21.575	21.575	3
Ge|Si 58	0.966	0.900	170.898	40.365	10
Sn|Si 23	0.978	0.897	333.921	102.330	10
Si|Si 16	0.956	0.846	1.251	1.147	10

3.3.1 Interface Rank Preservation

While absolute energy errors between MLP and DFT were large (e.g. RMSE ≈ 189 eV), they were secondary to the study's goal of rank preservation. In this context, MLPs effectively captured the energetic hierarchy of Ge and Sn interfaces.

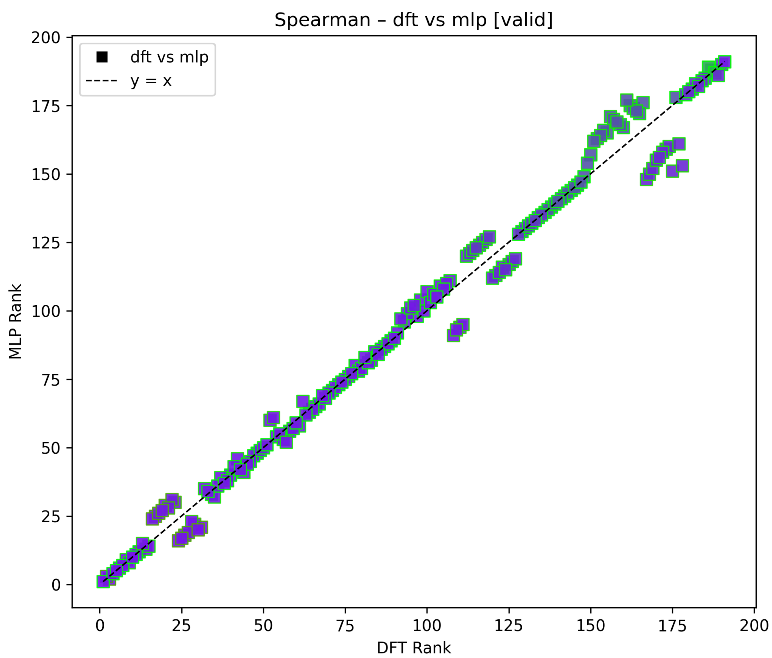

Figure 3.2: Spearman correlation for Ge interfaces.

Top-N overlap and rank correlation metrics confirmed that MLPs are useful for screening favourable structures, especially in Ge and Sn systems. Even when absolute energy values deviated significantly, the identity of the top-ranked configurations was largely preserved, with 8 out of 10 matches in many systems. This supports the use of MLPs like MACE in high-throughput pre-screening pipelines to prioritise candidates for costly DFT relaxation.

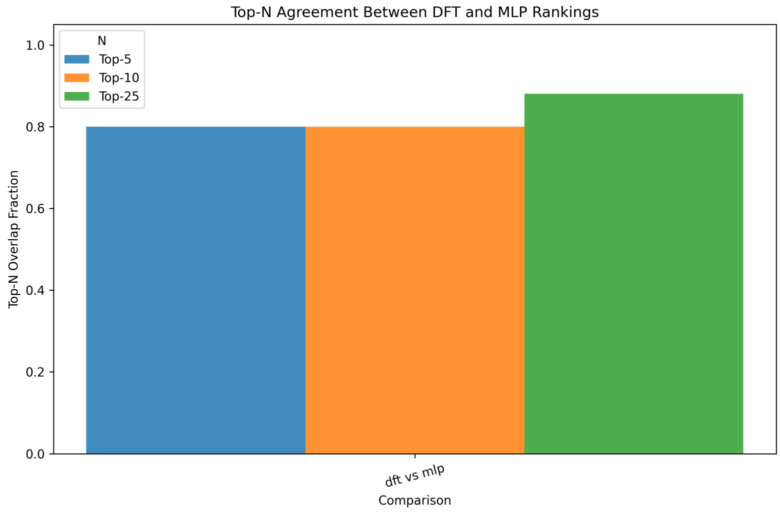

Figure 3.3: Top-*N* overlap for Si interfaces.

3.3.2 Generalisation and Limitations

MLPs offer significant computational speed advantages and can be reliably used for preliminary screening in systems with well-represented local bonding environments, such as Ge, Sn, and some $C|X$ combinations. However, their generalisation is limited in geometrically asymmetric or strained systems, such as $Si|Ge$ or $Sn|Si$, where stacking direction and coordination strongly influence prediction quality. Without targeted retraining or domain-specific corrections, MLP-based predictions in such cases may be misleading.

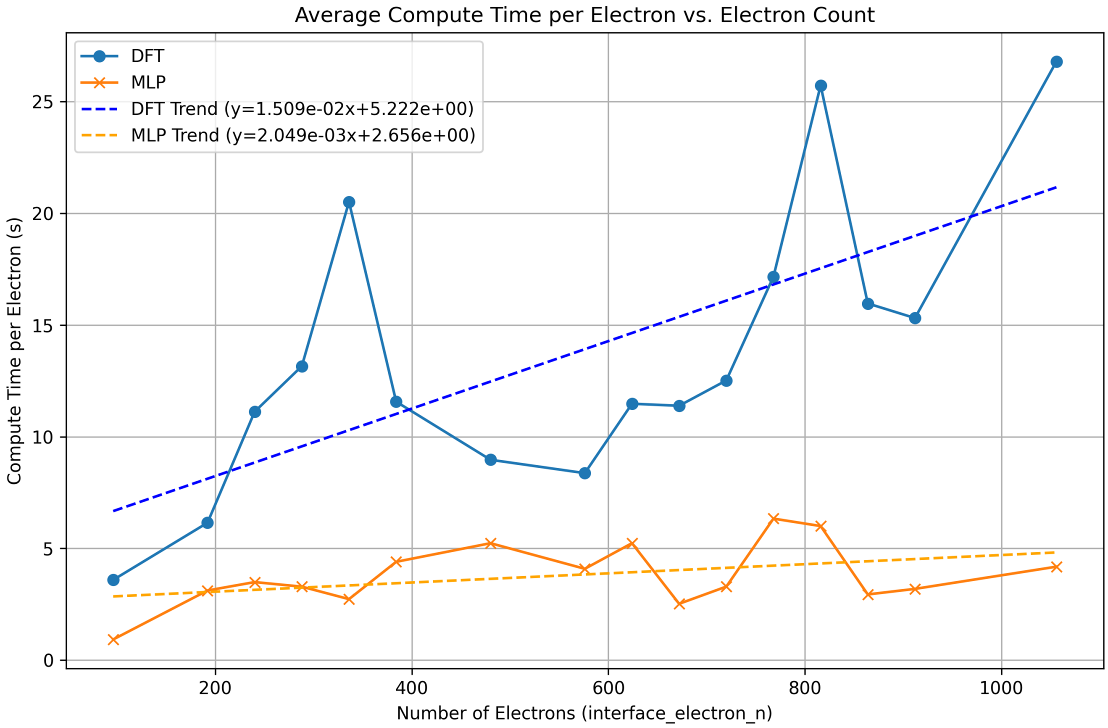

Figure 3.4: Average compute time per electron versus total electron count for interface structures, comparing DFT (blue) and MLP (orange) energy evaluations. Compute time was measured in walltime seconds on equivalent hardware. DFT exhibits a significantly steeper increase in cost with system size, with a linear trend slope of 1.51×10^{-2}, compared to just 2.05×10^{-3} for MLP—a roughly $7\times$ slower growth rate. This illustrates the much better scalability of machine-learned potentials for large-scale or high-throughput interface screening.

Chapter 4

Next Steps

4.1 Further Steps Towards Rank Preservation

The ability to preserve DFT-calculated interface rankings is critical for efficient high-throughput screening of material systems [17, 19, 36]. Absolute interface energies are less relevant in isolation if the relative favourability of competing configurations is misrepresented. This motivates the need for improving rank fidelity in machine-learned models such as MACE [11,19,29].

To address the limitations identified in Chapter 3, particularly in silicon-rich systems and mixed-coordination interfaces, future work will compare three MLP training strategies:

1. **Baseline MACE model** Trained primarily on bulk configurations.
   It offers good rank preservation in many systems but struggles in Si-rich or strained interfaces, where structural features deviate from the training distribution.

2. **Interface-specific fine-tuning** Enhances the baseline model using additional interfacial configurations.
   This aims to reduce domain mismatch and improve ranking accuracy in chemically or structurally diverse cases.

3. **Delta-learning correction** Learns the residual difference between MLP and

DFT energy outputs ($\Delta E = E_{\mathrm{DFT}} - E_{\mathrm{MLP}}$). This approach enables low-cost calibration without full retraining, making it attractive for incremental improvements [5].

To quantify the degree of rank preservation between MLP and DFT energy evaluations, we will use the following benchmarking metrics:

- **Spearman's rank correlation coefficient** (ρ): Measures how well the global ranking of structures is preserved between methods [49].
  Chapter 3 reports near-perfect Spearman values (e.g. $\rho = 0.9954$) in many systems.
- Kendall's rank correlation coefficient (τ): Assesses the proportion of concordant versus discordant ranking pairs. It is more sensitive to small local swaps in ranking than Spearman's ρ.
- Top-N Overlap: Evaluates how many of the top-ranked structures under DFT are also present in the MLP top-N list [25].
  This provides a practical indicator for screening performance.
- Mean Absolute Error (MAE) and Root Mean Square Error (RMSE): These absolute error metrics quantify the average and variance-weighted energy deviation, even when rank preservation is strong [5].

These metrics will be applied across chemically diverse systems (C, Si, Ge, Sn), and varied geometries including different Miller planes and stacking shifts. Structural asymmetry and slab orientation (upper vs lower slab) will also be included in the benchmarking matrix.

Figures such as Figure 4.2 and Figure 4.1 will illustrate performance nuances across system subsets.

Performance will be assessed using statistical correlation coefficients and overlap in Top-N rankings between DFT and MLP predictions. These metrics will provide fine-grained diagnostics of rank preservation .

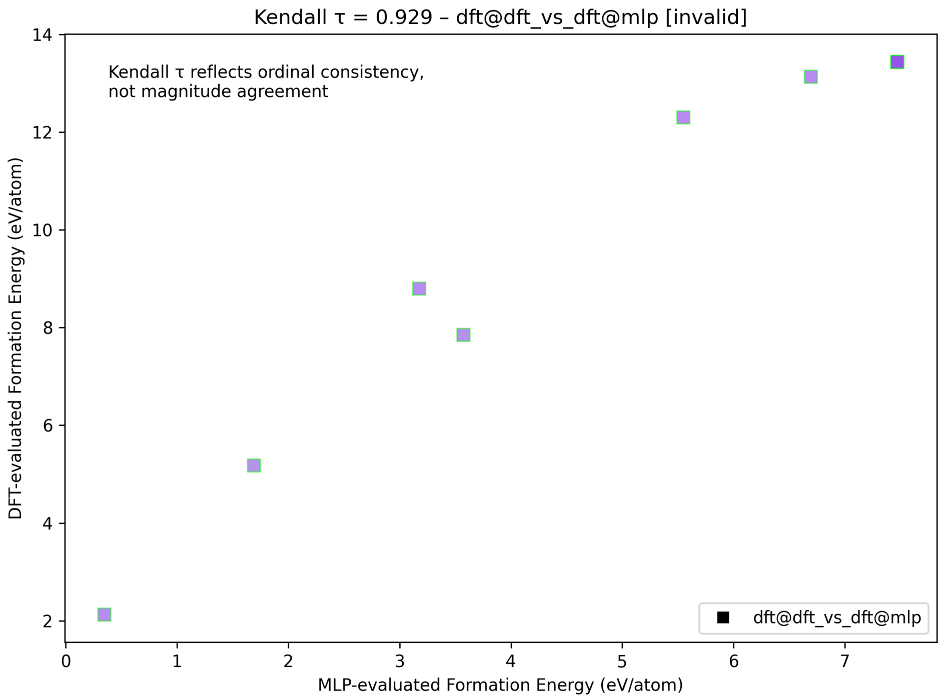

Figure 4.1: Example of rank degradation near a phase boundary: Kendall correlation for lower Ge / upper Sn interfaces. High deviation may indicate inversion.

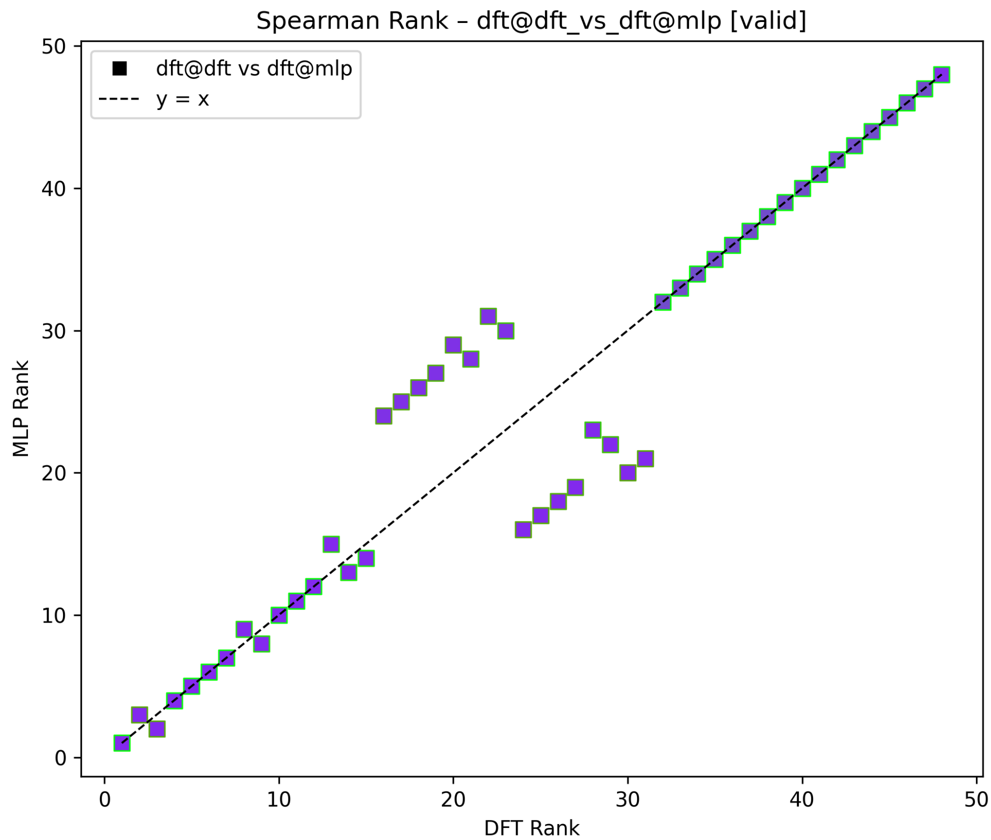

Figure 4.2: Grouped rank correlation performance by system and slab orientation. Example shown: Spearman ρ for Si in upper slab.

Table 4.1: Summary of benchmarking metrics used to compare MLP and DFT interface predictions.

Metric	Symbol	Interpretation
Spearman Rank Correlation	ρ	Preserves overall order of interface rankings
Kendall Rank Correlation	τ	Sensitive to local rank swaps; pairwise comparison
Top-N Overlap	–	Measures agreement in top-ranked structures
Mean Absolute Error	MAE	Average magnitude of energy deviation (eV)
Root Mean Square Error	RMSE	Penalises large deviations more heavily than MAE

Error Scaling with System Size

Preliminary results suggest that energy errors may scale with system size, particularly in Sn-rich heterostructures. For example, systems involving tin exhibited high RMSE values despite strong rank correlation ($\rho \approx 0.9744$). This may be due to larger atomic volumes, heavier elements, or deeper potential wells that amplify energy deviations in absolute terms. However, no systematic analysis of this scaling effect has yet been performed.

A robustness study will be conducted to quantify how MAE and RMSE evolve as a function of total atom count or interface area. If a clear trend emerges, this could motivate size-normalised corrections or domain-specific energy thresholds for MLP screening.

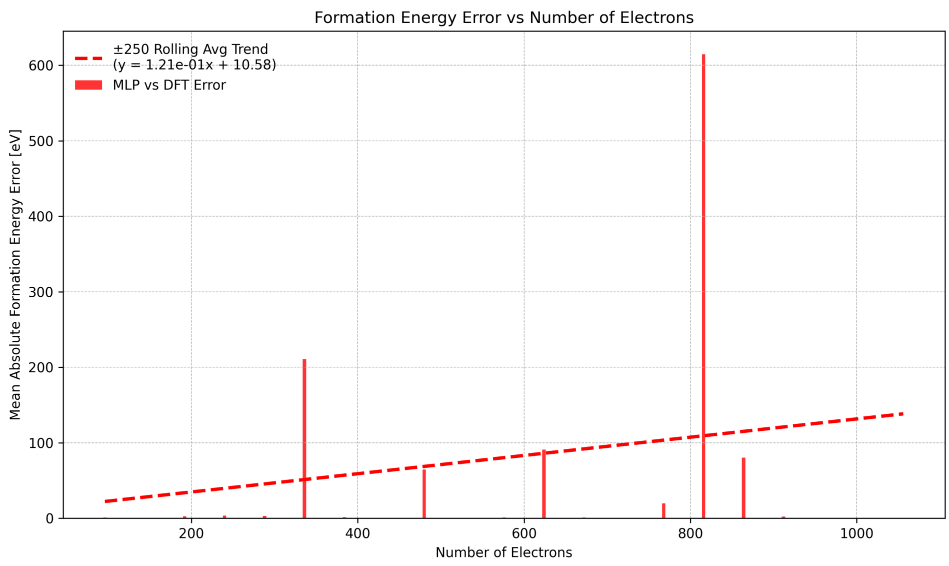

Figure 4.3: Mean absolute formation energy error between MLP and DFT predictions as a function of electron count for each interface structure. Each red bar represents the absolute error for a single structure, with a ± 250-electron rolling average trend (dashed red line) showing the overall error growth with system size. The fitted linear trend has a slope of -1.21×10^{-1}, indicating that despite fluctuations and a few outliers, the average error does not systematically increase with electron count. This suggests that MLP accuracy remains stable across a wide range of system sizes.

Tracking Failed Calculations and Dataset Completeness

Some interface evaluations, especially in large or unrealistic systems, failed due to relaxation non-convergence, broken geometries, or invalid energy outputs (e.g. $NaNs$). These were marked as `is_broken` and excluded from ranking analyses (see Chapter 3). However, no central mechanism currently exists to summarise failure rates or track completeness across chemical systems, orientations, or slab types.

Future iterations will include a completeness tracker that records:

- The number of generated vs successfully evaluated structures.
- Breakdown by chemical system (e.g. $Si|Ge$, $C|Sn$).

- Breakdown by slab orientation and stacking direction.
- Statistics on failure causes (e.g. relaxation timeout, unphysical overlaps).

This diagnostic layer will help identify whether certain systems are systematically excluded, potentially biasing the dataset or skewing model evaluation.

4.2 Challenges in Building an Accurate Interface Prediction Model

Despite the promising results demonstrated in Chapter 3, developing a machine-learned interatomic potential (MLIP) capable of reliably predicting interface stability remains a formidable challenge. Many failure modes arise from limitations in the training data, extrapolation behavior, bonding diversity, DFT reference fidelity, and intrinsic model architecture. This section outlines the key reasons why such models may fail, drawing on current literature and our own experience building and benchmarking MACE-based predictors.

A robust MLIP depends critically on the diversity, quantity, and physical relevance of its training set [6, 15, 57]. Our dataset, while spanning 680 heterostructures across four group 14 elements and 29 surfaces, may not fully capture the high-strain, defective, or chemically mixed environments found at real interfaces. Without configurations containing defects, alloy bonds (e.g. $Si|Sn$), or off-equilibrium geometries, the model risks systematic errors when encountering novel atomic arrangements. Moreover, data imbalance, such as overrepresentation of Si surfaces, can bias the model's predictions away from accurate generalisation.

Interfaces are inherently out-of-distribution cases, often involving strain, registry shifts, and mixed bonding environments not present in the training data [6, 19, 26, 44]. Even small extrapolations, such as a Si atom bonded to metallic $\beta - Sn$, can cause large errors if the MLIP has not seen similar local environments [6, 26, 46]. Moreover, if multiple interface configurations are close in energy, even minor inaccuracies can lead to rank inversions, undermining screening workflows [17, 19, 29].

Extending the model to handle semiconductors, metals, and insulators introduces qualitative differences in bonding: covalent, metallic, ionic, and hybrid regimes. A single MLIP may struggle to capture directional bonding in Si and delocalised metallic bonding in Sn, let alone polar surfaces or long-range interactions in ionic oxides [9, 37]. Many universal potentials underperform when stretched across these regimes, especially in predicting interfacial stability or reconstruction [15, 29].

MLIPs inherit the limitations of the DFT calculations they are trained on [15, 46, 57]. If hybrid functionals or relativistic corrections are omitted, the MLIP may learn inaccurate bonding energies, phase stabilities, or surface reconstructions, particularly for heavy elements like Pb or Au. Similarly, neglecting van der Waals forces can impair accuracy in weakly bonded or physisorbed interfaces. These issues limit transferability, especially in cross-class systems (e.g. semiconductor/metal or $2D/3D$ hybrid interfaces) [15, 29, 43, 46].

State-of-the-art models like MACE, M3GNet [8], and CHGNet have been shown to exhibit systematic underestimation of surface and defect energies, sometimes referred to as PES softening. This leads to overpredicted interface stability and can mislead ranking. Other known issues include poor uncertainty quantification, overfitting due to high model complexity, and numerical instability for large systems or heavy elements. Without mechanisms to detect extrapolation or assess prediction confidence, the model may silently fail [5, 15, 40].

Each of these failure modes represents a critical risk to model fidelity. Recognising them early provides guidance for mitigating strategies: expanding and diversifying training data, using active learning to discover critical configurations, incorporating physics-based corrections into the reference data, and applying uncertainty-aware predictions. While our current work demonstrates strong ranking performance in many systems, these broader challenges must be addressed before a general-purpose interface prediction model can be considered reliable.

4.3 Surface Effects on Interface Favourability

A key open question is whether the surface energetics of individual slabs can serve as predictors for the favourability of the resulting heterostructure. If a reliable correlation exists between surface energy and interface formation energy ΔE_{form}, it could significantly reduce the need for exhaustive interface enumeration [3, 9, 60, 61].

Future work will analyse the data in surface energies to quantify correlations between:

- The average surface energy of the two slabs.
- The minimum/maximum surface energy of the slab pair.
- The resulting interface energy ΔE_{form} from DFT and MLP evaluations.

This analysis will help determine whether high-energy surfaces are more reactive, and therefore more likely to form low-energy interfaces, or whether other factors such as registry and orientation dominate.

A figure plotting surface energy combinations against interface energies is planned to visualise this correlation. If strong trends are identified, slab energetics could be used as a lightweight proxy feature in the early stages of MLI pipeline screening.

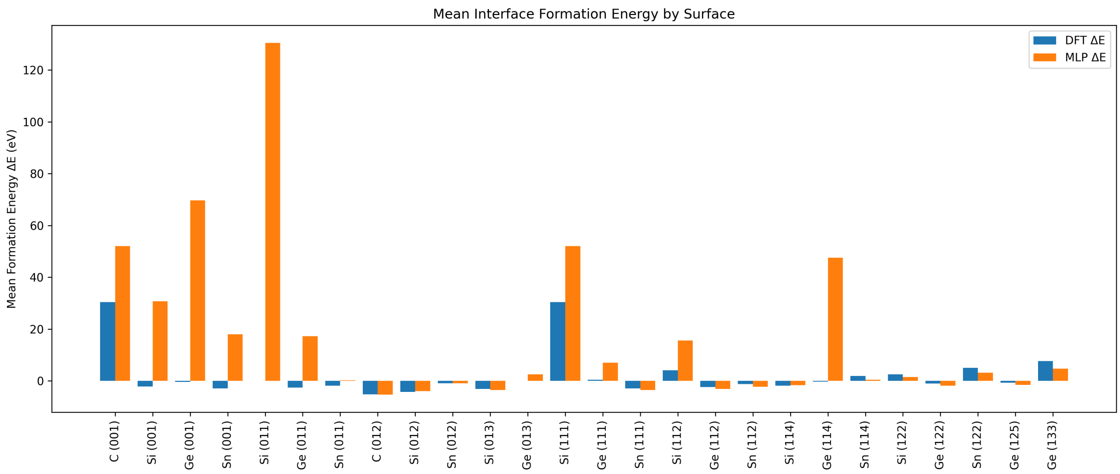

Figure 4.4: Mean interface formation energy (ΔE) by surface orientation and element, comparing DFT (blue) and MLP (orange) predictions. Each bar represents the average formation energy across all interfaces involving the specified surface (e.g. $Si(001)$, $Ge(111)$). While overall trends are consistent across methods, certain surfaces—such as $Si(011)$ and $Ge(114)$—show large discrepancies, highlighting cases where MLP overestimates formation energy relative to DFT. Orientations such as $Si(112)$ and $Ge(122)$ exhibit strong agreement. This analysis underscores the role of surface geometry in influencing interfacial stability and method-dependent deviations.

4.4 Crystallographic Structure and Interface Behaviour

Interface orientation and underlying crystal symmetry strongly influence both the physical plausibility and ranking fidelity of heterostructures [13]. Chapter 3 demonstrates that coherent interfaces with well-aligned Miller planes tend to exhibit better MLP-DFT rank agreement, particularly in systems with low strain and high registry.

Future work will group interfaces by:

- Parent symmetry (e.g. diamond vs. hexagonal structures)
- Orientation combinations (e.g. $\{100\}|\{100\}$, $\{100\}|\{111\}$)
- Slab asymmetry (e.g. Ge over Sn vs. Sn over Ge)

Metrics such as ρ, τ, and Top-N overlap will be aggregated for each group to identify structural contexts that promote or degrade predictive performance. For example, early results suggest that silicon in the lower slab often reduces rank fidelity compared to reversed configurations.

If resources permit, this analysis could be extended to other lattice types (e.g. bcc or hcp) to evaluate whether the observed trends generalise beyond diamond-structured group 14 systems.

4.5 Integration of RAFFLE

While ARTEMIS enables systematic construction of aligned interface stacks with controlled Miller plane matching and registry shifts, its deterministic approach limits structural diversity. This can lead to overrepresentation of high-symmetry configurations and under-sampling of disordered or experimentally relevant interfaces [51].

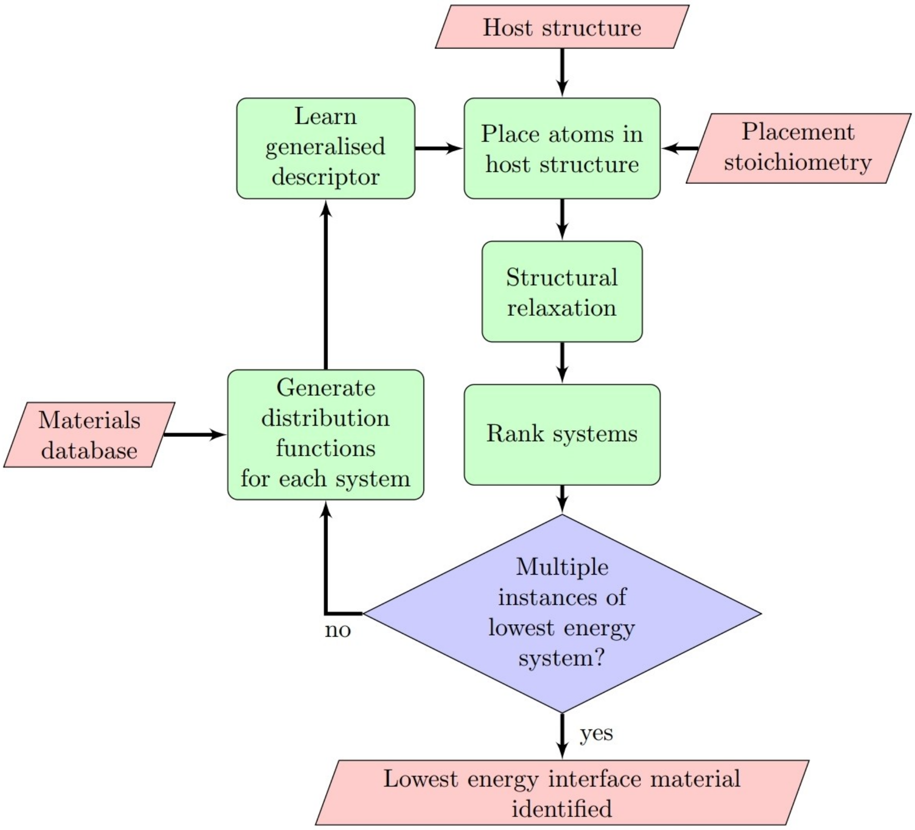

Figure 4.5: Overview of the RAFFLE workflow for interface structure prediction using active learning. The process begins with three user-defined inputs: a host structure, a target stoichiometry for atomic placement, and a reference database of previously evaluated systems. RAFFLE generates system-specific distribution functions, from which it learns generalised descriptors to guide atom placement within the host. Following structural relaxation, systems are ranked by energy. If multiple lowest-energy configurations are identified, the process converges. Otherwise, newly ranked systems are used to refine the descriptors and update placement strategies iteratively. Used with permission from Taylor et al. [39, 52].

RAFFLE addresses this limitation by introducing stochastic perturbations within the interfacial zone. Using the `raffle_generator` module, RAFFLE will:

- Vary atomic positions, stacking offsets, and coordination environments.
- Generate alloy-like bonding motifs and partial intermixing at the interface.
- Sample non-aligned registry patterns to mimic more realistic boundary configurations.

Initial slab structures and surface orientations will still be defined via ARTEMIS to ensure lattice coherence. RAFFLE will then apply controlled configurational noise or interpolation within the interface region, expanding the training space beyond aligned stacks.

This approach is especially useful for challenging systems such as Si|Sn and Ge|Sn, where perfect alignment is hard to achieve due to large lattice mismatch and differing atomic sizes. By broadening the configurational diversity, RAFFLE aims to:

- Improve generalisation of MLPs trained on interface data.
- Reveal edge cases where rank preservation fails due to disorder.
- Increase coverage of physically relevant but hard-to-discover interface motifs.

Comparative relaxation studies will be conducted using both DFT and MLP pipelines to assess the energetic spread and ranking fidelity introduced by RAFFLE. A schematic comparing ARTEMIS-only versus RAFFLE-augmented structure sets is planned to visualise the expansion of configurational space.

4.6 Machine-Learned Interface (MLI) Predictor

The ultimate goal of this pipeline is to develop a Machine-Learned Interface (MLI) predictor capable of estimating interface favourability directly from slab pair descriptors, without requiring full relaxation or enumeration of stacking shifts. This model will act as an early-stage filter to prioritise promising candidates for structure generation and DFT refinement.

Input features will include:

- Surface energies
- Crystallographic orientation identifiers (Miller indices)

- Lattice mismatch metrics (based on in-plane scaling ratios)
- Interface registry scores (from ARTEMIS or RAFFLE shifts)
- Atomic coordination and stoichiometry at the intended interface plane
- Analytical band alignment features (e.g. Anderson's rule offset and induced field correction)

Initial labels for training will be derived from interface formation energies, computed using either DFT or MACE total energies (following benchmarking). These energies quantify the relative stability of each structure compared to its bulk constituents, and are normalised per atom. Where available, training labels will be enriched with additional structural descriptors extracted from POSCAR files and relaxation trajectories.

Prediction targets will include:

- Binary classification: favourable vs. unfavourable interfaces.
- Regression: predicted interface formation energy ΔE_{int}.
- Rank scoring: predicted favourability percentile within a generated batch.

Simple models (e.g. decision trees, SVMs) will be used as baselines. If effective, graph-based architectures or equivariant neural networks may be explored to capture spatial features across slab boundaries.

The MLI predictor will integrate upstream with ARTEMIS and RAFFLE:

1. Given a pair of slabs, the MLI will score interface likelihoods before full stack generation.
2. High-scoring pairs will be passed to ARTEMIS for registry-matched stacking and to RAFFLE for disordered sampling.
3. Final relaxed energies will serve to update or retrain the MLI in an active-learning loop.

This workflow could enable a fully ML-driven interface discovery pipeline; one that uses minimal DFT data to bootstrap structure generation, ranking, and model refinement.

MLI Model Architecture and Software Perspective

The proposed Machine-Learned Interface (MLI) model will adopt a graph neural network (GNN) architecture, which is particularly well-suited to atomic-scale prediction tasks. In this formulation, atoms are represented as nodes, and edges encode interatomic relationships such as distance, bond type, or spatial proximity. Node features may include atomic number, electronegativity, or coordination number, while edge features capture pairwise distances, bond angles, or periodic adjacency. This graph-based abstraction mirrors the geometry and locality of real materials, enabling the model to learn chemically and physically meaningful patterns across interface structures.

GNNs are especially appropriate for this task because they respect the symmetries of atomistic systems; they are invariant or equivariant to translation, rotation, and permutation of atoms, depending on their architecture. Through message-passing and feature aggregation, each atom aggregates information from its neighbors to refine local descriptors, allowing the model to learn both local bonding patterns and emergent long-range effects. These properties are essential for predicting interfacial phenomena, which often arise from geometric asymmetry, registry mismatch, and complex bonding environments [6,15,46,48].

Implementation Pipeline and Integration

From a software engineering standpoint, the MLI model will be implemented as a modular, callable component in the interface discovery pipeline. It will accept structured slab-pair data; either parsed directly from POSCARs or preprocessed descriptor arrays, and produce predictions of interface favourability. The model will be constructed using scalable graph learning libraries such as PyTorch Geometric, DGL, or JAX-based GNN frameworks, and will provide a clean API for downstream tools such as ARTEMIS and RAFFLE. Outputs will be compatible with JSON or HDF5-based metadata systems, ensuring traceability and reproducibility.

Initial experiments may use relatively shallow GNNs with fixed neighbor cutoffs, but later iterations may incorporate equivariant tensor representations (e.g. MACE-style angular embeddings) to better capture orientation-dependent effects. Training targets will include both regression on ΔE_{form} and binary classification (favourable / unfavourable), optionally extended with uncertainty quantification. Crucially, the MLI is not meant to replace DFT or full relaxation pipelines, but to act as a high-speed prefilter that screen candidate structures for further evaluation. If successful, it will form the decision engine in a closed-loop ML-driven workflow for interfacial discovery.

Chapter 5

Concluding Thoughts

5.1 Insights from Work Undertaken

The carbon dataset exhibited strong performance when C appeared in the upper slab ($C|X$), with rank correlation $\rho \approx 0.95$ and low absolute error . However, only three $C|Si$ structures with carbon in the lower slab were usable, and these performed poorly, yielding negative rank correlation . As such, carbon results are informative but limited in generalisability .

Across all iterations, silicon-rich systems consistently showed poor agreement between MLP and DFT, particularly when Si was present in the lower slab . These failures were not isolated cases but appeared systematic, pointing to intrinsic limitations in the current MACE model for accurately representing Si bonding under strain or geometric asymmetry .

Ge and Sn systems, by contrast, showed robust agreement across all ranking metrics . These elements are likely well-covered in the MACE training distribution and may benefit from their more metallic bonding character, which simplifies energy regression .

Notably, $Si|Ge$ alloy systems exhibited near-perfect rank preservation, with both Spearman and Kendall values approaching 1.0 . These configurations involved uniform chemical environments and minimal interfacial disruption, suggesting that

MLPs perform best when atomic arrangements remain close to their bulk-trained regimes .

Table 5.1: Comparison of MLP performance on perfect alloys and heterostructures.

System	Spearman	Kendall	Top-10 Overlap
Perfect Alloys	1.00	1.00	6
Ge|Si Heterostructure	0.73	0.55	6

5.2 Broader Lessons and Limitations

MLPs appear reliable for ranking interfacial candidates when the bonding environments and geometries fall within the training domain . Screening workflows benefit most from MLPs in systems involving Ge, Sn, and well-registered C|X structures, where rank preservation is consistently strong . In contrast, strained or covalently dominated systems like Si|Ge or Si|C exhibit erratic performance, especially when interfacial bonding is asymmetric or poorly coordinated .

These results highlight the need for delta-correction strategies or domain-specific retraining, particularly for silicon-rich or mixed-interface systems . Hybrid workflows that combine MLP-based pre-screening with selective DFT confirmation offer a scalable and pragmatic approach to large-scale interface discovery .

This study reinforces the value of rank-based benchmarking; such as Spearman, Kendall, and Top-N overlap, over absolute energy metrics . In screening contexts, the ability to reliably identify top-performing configurations is more actionable than minimising raw energy error .

5.3 Concluding Remarks

This work demonstrates that while general-purpose MLPs like MACE can successfully preserve interface rankings in selected systems, their predictive utility is highly system- and geometry-dependent . Strong performance was observed in Ge, Sn, and $C|X$ systems with good atomic registry, as well as in chemically uniform $Si|Ge$ alloy configurations . In contrast, Si-rich and poorly coordinated interfaces exhibited systematic rank failures, underscoring the limits of current models in strained or covalently dominated regimes .

By enabling fast, large-scale screening of interfacial structures in favourable systems, this study lays the foundation for accelerated discovery workflows . However, widespread application across diverse material combinations will require targeted retraining, delta-learning, or integrated MLP-DFT hybrid schemes . Future work should continue to refine these approaches, using rank-preservation metrics as a primary benchmark for practical utility in materials design .

Despite the promising results in rank preservation for certain systems, this study has also highlighted substantial challenges in building a truly generalisable machine-learned potential for interface prediction . Limitations such as incomplete training coverage, extrapolation to novel configurations, and model underperformance in strained or covalently bonded systems underscore the need for further refinement . These risks are being actively addressed through targeted retraining, delta-learning corrections, and expanded benchmark testing, as outlined in Chapter 4. While strong performance was observed in Ge, Sn, and alloy systems, achieving consistent accuracy across a broader range of interfaces will require careful design of training data and model architecture .

Appendix A

Energy

Energy and Momentum in Classical and Quantum Physics: From Fundamentals to Material Models

Alexander Paul Holman

2025-04-06 UTC

A.1 Introduction

Understanding energy is fundamental to all of physics. From describing the motion of a single object to the behaviour of electrons in solids, energy forms a bridge between classical intuition and quantum mechanics. This document begins with classical kinetic energy, rewrites it in terms of momentum, and then builds toward a quantum mechanical framework. Finally, it expands on several important models used in solid-state physics to describe electron behaviour in materials.

A.2 Kinetic Energy in Classical Mechanics

In Newtonian physics, kinetic energy is the energy associated with motion:

$$E_{\text{kinetic}} = \frac{1}{2}mv^2 \tag{A.1}$$

where:

- E_{kinetic} is kinetic energy,
- m is mass,
- v is velocity.

This expression comes from the work-energy theorem. The quadratic dependence on velocity means that doubling the speed quadruples the kinetic energy.

A.3 Expressing Kinetic Energy in Terms of Momentum

Momentum is defined as:

$$p = mv \tag{A.2}$$

Rewriting kinetic energy:

$$v = \frac{p}{m} \tag{A.3}$$

$$E = \frac{1}{2}mv^2 = \frac{1}{2}m\left(\frac{p}{m}\right)^2 = \frac{p^2}{2m} \tag{A.4}$$

This form is particularly useful in quantum mechanics, where momentum becomes more fundamental than velocity.

A.4 Quantum Mechanical Momentum and Energy

In quantum mechanics, particles exhibit wave-particle duality. According to de Broglie's hypothesis:

$$p = \hbar k \tag{A.5}$$

where $\hbar$ is the reduced Planck constant, and k is the wave vector.

Substituting into the kinetic energy:

$$E_{\text{kinetic}} = \frac{p^2}{2m} = \frac{(\hbar k)^2}{2m} = \frac{\hbar^2 k^2}{2m} \tag{A.6}$$

SPH has added for discussion

$$\hat{p} = i\hbar\nabla \tag{A.7}$$

This is the kinetic energy of a free quantum particle.

A.5 Total Energy in Quantum Systems

In real systems, kinetic energy alone is not sufficient. The total energy includes interactions and external fields:

$$E_{\text{total}} = E_{\text{kinetic}} + E_{\text{potential}} \tag{A.8}$$

A.5.1 Interactions and Potentials

Let us define:

- E_{i-i}: ion–ion interactions
- E_{e-e}: electron–electron interactions
- E_{i-e}: ion–electron interactions

Each is expressed as:

$$E_\alpha = \sum_{i,j} V_\alpha(|\mathbf{r}_i - \mathbf{r}_j|), \quad \alpha \in \{\text{i-i}, \text{e-e}, \text{i-e}\} \tag{A.9}$$

Total potential energy:

$$E_{\text{potential}} = \sum_{\alpha \in S} \sum_{i,j} V_\alpha(|\mathbf{r}_i - \mathbf{r}_j|) \tag{A.10}$$

A.6 Models of Electrons in Solids

Many-body quantum systems are challenging to solve exactly, so simplified models are used.

A.6.1 Free Electron Model (Drude–Sommerfeld)

Assumptions:

- Electrons are non-interacting
- No periodic potential
- Infinite potential walls

$$E = \frac{\hbar^2 k^2}{2m} \tag{A.11}$$

Leads to parabolic dispersion:

- Explains electrical conductivity
- Cannot explain band gaps or semiconductors

A.6.2 Nearly Free Electron Model

Adds a weak periodic potential:

$$\hat{H} = \frac{\hat{p}^2}{2m} + V_{\text{periodic}}(\mathbf{r}) \tag{A.12}$$

- Band gaps emerge at Brillouin zone boundaries
- Explains metallic behaviour and electronic bands

A.6.3 Tight-Binding Model

Electrons are localized around atoms and hop between them:

$$\hat{H} = \sum_i \epsilon_i \hat{c}_i^\dagger \hat{c}_i + \sum_{\langle i,j \rangle} t_{ij} \hat{c}_i^\dagger \hat{c}_j \tag{A.13}$$

Where:

- ϵ_i is the on-site energy
- t_{ij} is the hopping parameter
- $\hat{c}_i^\dagger$, $\hat{c}_i$ are creation/annihilation operators

Suitable for insulators and semiconductors.

A.6.4 Effective Mass Approximation

Electrons near band extrema behave like free particles with effective mass:

$$E(k) \approx E_0 + \frac{\hbar^2 k^2}{2m^*} \tag{A.14}$$

Effective mass is defined as:

$$\frac{1}{m^*} = \frac{1}{\hbar^2} \frac{d^2 E}{dk^2} \tag{A.15}$$

A.6.5 Density Functional Theory (DFT)

DFT reformulates the many-body problem in terms of electron density:

$$E[n] = T_s[n] + \int V_{\text{ext}}(\mathbf{r}) n(\mathbf{r})\, d\mathbf{r} + E_H[n] + E_{\text{xc}}[n] \tag{A.16}$$

Where:

- $T_s[n]$: Kinetic energy of non-interacting electrons
- $E_H[n]$: Hartree energy
- $E_{\text{xc}}[n]$: Exchange-correlation energy

Used for computing total energies, geometries, and electronic structure.

A.7 Summary and Perspective

We have the following:

- Transitioned from classical to quantum formulations
- Built from simple to complex models: Free Electron, Nearly Free, Tight-Binding, Effective Mass, and DFT
- Seen how abstraction enables accurate and scalable modelling

Understanding these transitions is essential for theoretical and computational physicists. It allows for the appropriate selection of models across scales and materials, laying a foundation for exploring conductivity, band theory, and phase transitions.

Appendix B

Forces

Forces in Classical and Quantum Physics:
Foundations and Extensions Alexander Paul Holman 2025-04-06 UTC

B.1 Introduction

Forces are the drivers of motion and interaction. While energy describes what is possible, force describes what is happening. In this document, we explore force from multiple perspectives—Newtonian, potential-based, field-theoretic, quantum mechanical, and within density functional theory (DFT). We explain how forces emerge, how they relate to energy, and how they behave in physical systems.

B.2 Force in Newtonian Mechanics

The concept of force in classical mechanics is governed by Newton's Second Law:

$$\mathbf{F} = m\mathbf{a} \tag{B.1}$$

Since acceleration is the second derivative of position:

$$\mathbf{F} = m\frac{d^2\mathbf{r}}{dt^2} \tag{B.2}$$

This formulation underlies classical mechanics and describes macroscopic motion.

B.3 Force as the Gradient of Potential Energy

In conservative systems, forces arise from potential energy fields:

$$\mathbf{F} = -\nabla V(\mathbf{r}) \tag{B.3}$$

Examples include:

- Gravity: $V(z) = mgz \Rightarrow \mathbf{F} = -mg\hat{z}$
- Harmonic oscillator: $V(x) = \frac{1}{2}kx^2 \Rightarrow F = -kx$
- Coulomb force: $V(r) = \frac{1}{4\pi\varepsilon_0}\frac{q_1q_2}{r} \Rightarrow \mathbf{F} = -\frac{dV}{dr}\hat{r}$

B.4 Types of Classical Forces

B.4.1 Contact Forces

Normal, friction, tension, spring:

$$F_{\text{spring}} = -kx \tag{B.4}$$

B.4.2 Field Forces

- Gravitational: $F = G\frac{m_1m_2}{r^2}$
- Electrostatic: $F = \frac{1}{4\pi\varepsilon_0}\frac{q_1q_2}{r^2}$
- Magnetic: $\mathbf{F} = q\mathbf{v} \times \mathbf{B}$

B.5 Lagrangian and Hamiltonian Mechanics

B.5.1 Lagrangian Formulation

$$L = T - V, \quad \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}}\right) - \frac{\partial L}{\partial q} = 0 \tag{B.5}$$

Force appears through the variation of the action and generalized coordinates.

B.5.2 Hamiltonian Formulation

$$\frac{dq}{dt} = \frac{\partial H}{\partial p}, \tag{B.6}$$

$$\frac{dp}{dt} = -\frac{\partial H}{\partial q} \tag{B.7}$$

Force is related to the rate of momentum change in phase space.

B.6 Quantum Mechanical Force

B.6.1 Ehrenfest's Theorem

$$\frac{d}{dt}\langle \hat{p} \rangle = \langle -\nabla V(\hat{r}) \rangle \tag{B.8}$$

This connects classical and quantum descriptions through expectation values.

B.6.2 Operator View

Momentum is an operator:

$$\hat{p} = -i\hbar\nabla \tag{B.9}$$

and force derives from potential gradients acting on wavefunctions.

B.7 Force Fields in Classical and Quantum Physics

B.7.1 Classical Fields

- Electric: $\mathbf{E} = -\nabla V$
- Gravitational: $\mathbf{g} = -\nabla \Phi$
- Lorentz force: $\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$

B.7.2 Quantum Fields

In quantum field theory, forces arise from particle exchange:

- Electromagnetism $\rightarrow$ photons
- Strong force $\rightarrow$ gluons
- Weak force $\rightarrow$ W/Z bosons
- Hypothetical gravity $\rightarrow$ gravitons

B.8 Forces in Materials and Solids

Atomic interactions govern material structure and behavior. Interatomic potentials model forces:

$$\mathbf{F}_i = -\nabla_i V(\{\mathbf{r}_j\}) \tag{B.10}$$

Used in:

- Molecular dynamics
- Elasticity and plasticity
- Phonon and vibrational analyses

B.9 Forces in Density Functional Theory (DFT)

DFT allows quantum-mechanical force calculations on atoms within a material.

B.9.1 Hellmann–Feynman Theorem

$$\mathbf{F}_I = -\frac{dE_{\text{tot}}}{d\mathbf{R}_I} = -\left\langle \psi \left| \frac{d\hat{H}}{d\mathbf{R}_I} \right| \psi \right\rangle \tag{B.11}$$

As long as the wavefunction ψ is self-consistent, this expression gives accurate forces without needing derivatives of ψ.

B.9.2 Force Contributions

Total force in DFT codes includes:

- Hellmann–Feynman force
- Pulay force (basis set dependent)
- Non-local pseudopotential terms

$$\mathbf{F}_I = \mathbf{F}_I^{\text{HF}} + \mathbf{F}_I^{\text{Pulay}} + \mathbf{F}_I^{\text{NL}} \tag{B.12}$$

B.9.3 Applications in DFT

- Geometry optimization: relaxing structures to minimum energy
- Phonon calculations: displacing atoms to find vibrational modes
- Ab initio molecular dynamics (AIMD): time-evolving systems using DFT-computed forces

B.9.4 Classical Link via Born–Oppenheimer Approximation

Although forces are quantum-derived, the atomic motion follows classical mechanics:

$$m_I \frac{d^2\mathbf{R}_I}{dt^2} = \mathbf{F}_I^{\text{DFT}} \tag{B.13}$$

This duality is the cornerstone of quantum simulations in condensed matter physics.

B.10 Further Discussion and Applications

B.10.1 Implications of Quantum Forces on Macroscopic Physics

Emergence of Classical Behavior from Quantum Forces

- **Material Properties**: Mechanical strength, elasticity, thermal conductivity, and optical characteristics are all rooted in quantum-level forces.
- **Phase Behavior**: Collective phenomena like melting and superconductivity arise from interactions such as van der Waals and exchange forces.
- **Chemical Reactions**: Governed by quantum potential energy surfaces (PES), affecting pathways and reaction rates.

Quantum Effects That Persist at Macroscopic Scales

- **Superconductivity**, **Quantum Hall Effect**, **Casimir Effect**

Implications in Engineering and Technology

- **Semiconductors**, **Nanotechnology**, **Quantum Sensors and Metamaterials**

B.10.2 Comparison of Interatomic Potentials

Potential Type	Key Features	Suitable For	Limitations
Lennard-Jones	Simple 2-body	Noble gases	Not directional or reactive
Morse	Bond forming/breaking	Diatomics	Not for complex systems
EAM	Many-body, electron density	Metals	Not covalent/ionic system
Tersoff/Brenner	Bond-order, angle-based	Semiconductors	Parameter sensitive
ReaxFF	Reactive, dynamic bonds	Complex chemistry	High cost, empirical
ML Potentials	Trained on DFT	General-purpose	Data hungry

Material-Specific Selection

- **Metals**: EAM, ML
- **Oxides/Ceramics**: ML, Buckingham, DFT
- **Semiconductors**: Tersoff, ML
- **Organics**: ReaxFF, ML
- **Interfaces**: DFT or hybrid ML

B.10.3 Experimental Validation Techniques

Structural Validation

- XRD, Neutron Diffraction, Electron Microscopy

Vibrational Validation

- Raman, IR, Neutron Scattering, Ultrafast Spectroscopy

Electronic and Force Response

- ARPES, STM/STS, Compton Scattering

Mechanical and Thermodynamic

- Nanoindentation, Thermal Conductivity, Elastic Moduli

Direct Force Mapping

- AFM, Non-contact AFM

B.11 Summary

Force unifies mechanics, quantum theory, and materials modeling. Whether it acts through fields, emerges from potentials, or is computed from quantum states, force remains central to our ability to describe, predict, and manipulate physical systems.

In classical physics, it accelerates masses. In quantum systems, it shifts expectation values and influences wavefunctions. In DFT, it allows us to discover stable materials and simulate atomic motion with quantum precision.

Appendix C

Crystal Structure

Crystal Structures in Two and Three Dimensions:
Lattices, Space Groups, and Modelling Applications Alexander Paul Holman 2025-04-06 UTC

C.1 Introduction

Crystalline materials are defined by the periodic arrangement of atoms in space. These arrangements can occur in two or three dimensions depending on the physical or computational constraints of the system. This document provides a structured overview of 2D and 3D crystal structures, including lattice systems, Bravais lattices, space groups, and their relevance to materials modelling.

C.2 Two-Dimensional (2D) Crystal Structures

C.2.1 2D Bravais Lattices

There are five distinct 2D Bravais lattices, each defined by its symmetry and unit cell geometry:

Lattice Type	Geometry	Angle Constraints	Examples
Oblique	Parallelogram	Arbitrary	Strained materials
Rectangular	Rectangle	$90°$	Black phosphorus
Centered Rectangular	Rectangle w/ center	$90°$	Modified 2D lattices
Square	Square	$90°$, equal sides	Monolayer Pb, 2D perovs
Hexagonal	Rhombus	$60°/ 120°$, equal sides	Graphene, h-BN, MoS_2

C.2.2 Notable 2D Materials

- **Graphene:** Hexagonal, Dirac cones, zero band gap
- **h-BN:** Hexagonal, wide-gap insulator
- **MoS_2:** Hexagonal, semiconducting TMD
- **Phosphorene:** Rectangular, anisotropic, high mobility
- **Silicene/Germanene:** Hexagonal, buckled graphene analogues

C.3 Three-Dimensional (3D) Crystal Structures

C.3.1 Lattice Systems

Seven lattice systems classify 3D unit cell geometries:

System	Axes	Angles
Cubic	$a = b = c$	$\alpha = \beta = \gamma = 90°$
Tetragonal	$a = b \neq c$	$\alpha = \beta = \gamma = 90°$
Orthorhombic	$a \neq b \neq c$	$\alpha = \beta = \gamma = 90°$
Monoclinic	$a \neq b \neq c$	$\alpha = \gamma = 90°,\ \beta \neq 90°$
Triclinic	$a \neq b \neq c$	$\alpha \neq \beta \neq \gamma \neq 90°$
Hexagonal	$a = b \neq c$	$\alpha = \beta = 90°,\ \gamma = 120°$
Rhombohedral	$a = b = c$	$\alpha = \beta = \gamma \neq 90°$

C.3.2 Bravais Lattices

Combining lattice systems with centering types gives 14 unique Bravais lattices:

- **Primitive (P):** Lattice points at corners
- **Body-centered (I):** Extra point at center
- **Face-centered (F):** Points at all faces
- **Base-centered (C/A/B):** Points at one pair of faces

Examples:

- Simple Cubic (P)
- BCC (Body-Centered Cubic)
- FCC (Face-Centered Cubic)
- HCP (Hexagonal Close-Packed) – not a Bravais lattice

C.4 Crystal Structure Examples

Material	System	Bravais	Notes
Silicon	Cubic	FCC-based (Diamond)	Covalent bonding
Iron	Cubic	BCC	Room temperature phase
Copper	Cubic	FCC	High conductivity
Rutile TiO_2	Tetragonal	Primitive	Oxide semiconductor
Calcite	Trigonal	Rhombohedral	Carbonate mineral
Quartz	Trigonal	Primitive	Piezoelectric

C.5 Space Groups: The Language of Crystal Symmetry

C.5.1 What Is a Space Group?

A space group defines all symmetry operations of a crystal:

- Translations
- Rotations
- Reflections
- Inversions
- Glide planes
- Screw axes

Space groups combine point groups with translational symmetry.

C.5.2 Classification and Notation

Dimension	**Count**	**Notes**
2D	17	Wallpaper groups
3D	230	Based on 32 point groups and 14 Bravais lattices

Common symbols (Hermann–Mauguin):

- P1: Only translations
- Fm$\bar{3}$m: FCC with cubic symmetry (e.g. NaCl)
- P6_3/mmc: HCP (e.g. Mg, Ti)

C.5.3 Examples

Material	System	Group #	Symbol
Silicon	Cubic	227	$Fd\bar{3}m$
Diamond	Cubic	227	$Fd\bar{3}m$
NaCl	Cubic	225	$Fm\bar{3}m$
Graphite	Hexagonal	194	$P6_3/mmc$
Rutile TiO_2	Tetragonal	136	$P4_2/mnm$
MoS_2	Hexagonal	194	$P6_3/mmc$
Quartz	Trigonal	152	$P3_121$ or $P3_221$

C.5.4 Importance in Modelling

- **DFT:** Space groups reduce k-point sets, saving computation
- **ML:** Encoded as features for symmetry-aware learning
- **Band structure:** Affects degeneracies and selection rules
- **Interfaces:** Governs strain and defect behaviour

C.5.5 Tools and Software

- **VESTA:** Visualization, symmetry operations
- **Spglib:** Symmetry detection (e.g., in ASE, pymatgen)
- **FINDSYM:** Online space group tool
- **Materials Project** / **AFLOW** / **NOMAD:** Symmetry-rich databases

C.6 Summary

This document provided a structural overview of crystal types, from 2D and 3D Bravais lattices to space groups. These symmetries underpin modern approaches in density functional theory, interface modelling, and machine learning. The mastery of the lattice and symmetry concepts is essential for accurate materials modelling.

Appendix D

Problem Space

Problem Space Alexander Paul Holman 2025-054-13 UTC

Core Points

1. Limits of Schrödinger's Equation

- Schrödinger's equation is non-relativistic and breaks down for heavy elements ($Z > 83$).
- Fails to capture spin-orbit coupling, relativistic mass effects, and orbital shifts.
- Inaccurate for elements like bismuth, uranium, etc., where core electrons are relativistic.

The Schrödinger equation forms the basis of non-relativistic quantum mechanics and underpins many standard approaches to electronic structure theory, including density functional theory (DFT). However, it assumes that particles move significantly slower than the speed of light and that relativistic effects can be neglected. This assumption breaks down for elements with high atomic numbers, particularly in the region of bismuth ($Z = 83$) and beyond. In these systems, the inner electrons experience intense electric fields and velocities approaching relativistic limits, invalidating Schrödinger-based solutions.

While the Schrödinger equation is a good approximation for light elements and remains effective across a wide portion of the periodic table, it fails to capture critical relativistic phenomena such as spin-orbit coupling, relativistic mass increase, and the contraction or expansion of orbitals. As a result, it becomes necessary to introduce fully relativistic formulations to describe the behavior of electrons in such environments accurately.

2. Relativistic Quantum Mechanics

- **Dirac equation** handles spin-½ particles (e.g., electrons) and includes relativity.
- **Klein-Gordon equation** handles spin-0 bosons but is incompatible with Dirac for mixed systems.
- Action-based formalisms (e.g., Lagrangian mechanics) are more general but computationally costly.

To address the breakdown of non-relativistic quantum mechanics, the Dirac equation was introduced as a relativistic extension suitable for spin-½ particles such as electrons. It inherently incorporates spin, predicts the existence of antimatter, and provides corrections to energy levels observable in high-resolution spectroscopy. A related alternative, the Klein-Gordon equation, describes spin-0 bosons but is not applicable to electrons.

However, these equations are not mutually compatible for systems containing both fermions and bosons, and they do not account for interactions between multiple particles or between particles and fields. As a result, high-fidelity modeling often relies on **action-based formulations** derived from quantum field theory. These treat the system through the principle of least action and provide a more general description capable of unifying disparate particle types and interactions.

In practice, such approaches are extremely computationally expensive and difficult to converge. As a result, condensed matter and materials physics frequently resort to semi-relativistic corrections or perturbative techniques, especially within DFT frameworks.

3. DFT Challenges for Heavy Elements

- Standard DFT errors often stem more from poor functionals than missing relativity.
- For heavy elements, **relativistic corrections** (e.g., spin-orbit coupling) become essential.
- Fully relativistic DFT exists but is slower, harder to converge, and lacks standard benchmarks.

Density functional theory is widely used for modeling materials across physics, chemistry, and engineering. However, its accuracy hinges on two primary aspects: (1) the functional form of the exchange-correlation energy, and (2) the treatment of relativistic effects. For many systems, errors introduced by an approximate functional outweigh those from neglecting relativity. However, as one moves toward heavier elements, **relativistic corrections can dominate** and must be incorporated.

These are typically introduced via:

- **Relativistic pseudopotentials**, which incorporate mass-velocity and Darwin terms.
- **Spin-orbit coupling**, either treated perturbatively or included directly in the Hamiltonian.
- **Fully relativistic Kohn-Sham equations**, solved using either two-component or four-component wavefunctions.

Despite their importance, these methods suffer from practical limitations. Spin-orbit-coupled calculations are significantly more expensive and notoriously difficult to converge. Furthermore, there are very few standardized benchmarks or robust functionals specifically validated for heavy-element systems, making verification and validation more difficult.

4. Pseudopotentials and Mixed-Row Interfaces

- Pseudopotentials simplify simulations by masking core electrons but:
  - Are unreliable for heavy atoms or where core-valence interactions matter.
  - Cause problems in **mixed-row systems** like Si|Ge or Sn|Pb interfaces.
- May require **full-electron calculations** for accurate interface modeling—expensive but necessary.

A further complication arises from the use of **pseudopotentials**, which are employed to reduce computational load by eliminating core electrons from the simulation. While effective for light and mid-period elements, pseudopotentials can be highly unreliable for systems where:

- The **core electrons contribute** directly to bonding (as in late transition metals).
- **Relativistic effects influence core-core and core-valence interactions**.
- **Mixed-row systems** (e.g., Si and Ge) introduce inconsistencies in pseudopotential transferability.

In particular, at interfaces between elements from different rows of the periodic table, pseudopotential mismatch can lead to unphysical artifacts such as incorrect band alignment or spurious states at the interface. These issues are exacerbated when the system lies on the borderline of relativistic significance—where the simplifications made in one component (e.g., a pseudopotential for Si) are invalid for the other (e.g., Ge or Sn).

The result is that **full-electron calculations** may be necessary, even for elements where they are usually avoided, in order to capture interfacial effects accurately. However, these calculations come at considerable computational cost and require careful convergence testing, often without robust tools for automation or standardization.

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